

Fundamentals of machine learning

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Probability of error and decision boundaries

Training data

- The "standard" process for obtaining data for a task is as follows:

→ Obtain x_i from an initial deployment / available data $x_i \sim P(x)$

→ Obtain label y_i manually / from outcomes $y_i \sim P(w | x_i)$

→ these are "draws" from the joint dist $p(w, x)$

- This yields supervised training data

$$D = \{(x_1, y_1), \dots, (x_N, y_N)\}$$

where x_i is the observation / feature vector

and $y_i \in \{w_1, \dots, w_k\}$ is the class label for observation x_i

Nature of classifier

- Classifiers can be broadly split as

① Generative models $p(x, w; \theta)$ ← we could use this to generate synthetic data

→ Model of the joint dist. of observations and classes is trained

→ Posterior dist. of class w_i obtained using Bayes rule:

② Discriminative models $p(w | x; \theta)$ ← cannot generate synthetic data unless given x

→ Model the posterior dist. of the class given the observations is trained

③ Discriminant functions

→ A mapping from an observation x to a class w is directly trained

→ No posterior probability generated, just class label.

Loss functions

- A loss function is a mathematical way to measure how good / bad the predictions of a ML method are compared to the actual values in the data.

- common loss functions include:

→ Squared error

$$L(\hat{w}, w) = (\hat{w} - w)^2$$

→ 0-1 loss

$$L(\hat{w}, w) = \begin{cases} 0 & \text{if } \hat{w} = w \\ 1 & \text{o/w} \end{cases}$$

→ Cross entropy

$$L(P, w) = - \sum_i w_i \log p_i$$

P is probability vector,
 w is one-hot encoding vector

* Loss functions may be unequal for particular applications

$$\rightarrow \text{eg: } L(f(x; \theta), w_1) = \begin{cases} 0 & f(x; \theta) = w_1 \\ c_1 & f(x; \theta) = w_2 \end{cases}$$

$$L(f(x; \theta), w_2) = \begin{cases} c_2 & f(x; \theta) = w_1 \\ 0 & f(x; \theta) = w_2 \end{cases} \quad \text{where } c_1 \neq c_2$$

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Expected loss, empirical loss and holdout empirical loss

- We can obtain a good predictor function $f(x; \theta)$ by training the model parameters θ to minimise the expected loss L_{act} , given by ^{average loss over K classes} ^{average over observations x}

$$L_{\text{act}} = \int \left[\sum_{i=1}^K L(f(x; \theta), w_i) P(w_i | x) \right] p(x) dx$$

where $P(w_i | x)$ is the "true" prob. of class given observation
and $p(x)$ is the "true" prob. of an observation

neither of these is usually known

- We can instead compute the empirical loss L_{emp} from the training data D , which are samples drawn from $p(w, x) = P(w | x) p(x)$.

$$L_{\text{emp}} = \frac{1}{N} \sum_{i=1}^N L(f(x_i, \theta), y_i)$$

by Monte Carlo, $\lim_{N \rightarrow \infty} L_{\text{emp}} = L_{\text{act}}$, but in practice N is finite so $L_{\text{emp}} \leq L_{\text{act}}$

- However, we only care about holdout data performance, so just consider the holdout evaluation set of N^* samples to compute the holdout empirical loss L_{eval}

$$L_{\text{eval}} = \frac{1}{N^*} \sum_{i=1}^{N^*} L(f(x_i^*, \theta), y_i^*)$$

note loss function used for training can be diff. to those used to compute L_{eval} .

Bayes decision rule

- Given an i.i.p x^* , we need to make a "decision" $\hat{w} \in \{w_1, \dots, w_K\}$, and associated w/ any decision is a loss/risk $L(\hat{w}, y^*)$, where $y^* \in \{w_1, \dots, w_K\}$ is the "correct" outcome.
- * Due to uncertainty, $y^* \sim P(w_i | x^*)$ ("true" prob. of class given observation, unknown)
- Bayes's decision rule minimises the expected loss to give the optimal decision

$$\hat{w} = \underset{w}{\text{argmin}} \left\{ \sum_{i=1}^K L(w, w_i) P(w_i | x^*) \right\}$$

and we train a model $P(w | x^*; \theta)$ to approx. unknown "true" prob. $P(w_i | x^*)$

- For equal losses across classes, we select the class w/ highest posterior $P(w | x^*; \theta)$,

$$\hat{w} = f(x^*; \theta) = \underset{w}{\text{argmax}} P(w | x^*; \theta) \approx \underset{w}{\text{argmax}} p(w | x^*)$$

Probability of error

- For 0-1 loss function, we can estimate the prob. of error w/ the holdout empirical loss L_{eval} ,

$$p(\text{error}) \approx L_{\text{eval}} = \frac{1}{N^*} \sum_{i=1}^{N^*} L(f(x_i^*, \theta), y_i)$$

- For a classifier $f(x; \theta)$ that yields two regions $\Omega_1 \rightarrow w_1$, $\Omega_2 \rightarrow w_2$, the prob. of error is

$$p(\text{error}) = P(w_1, x \in \Omega_2) + P(w_2, x \in \Omega_1) = \int_{\Omega_2} p(w_1, x) dx + \int_{\Omega_1} p(w_2, x) dx$$

This can be expressed in two forms

$$\rightarrow \text{Generative model: } p(\text{error}) = \int_{\Omega_2} p(x | w_1) P(w_1) dx + \int_{\Omega_1} p(x | w_2) P(w_2) dx$$

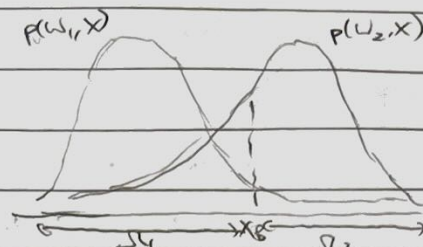
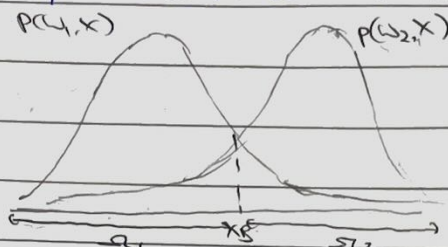
$$\rightarrow \text{Discriminative model: } p(\text{error}) = \int_{\Omega_2} P(w_1 | x) p(x) dx + \int_{\Omega_1} P(w_2 | x) p(x) dx$$

- The optimal decision boundary x_0 is where the min. prob. of error is obtained (same decision boundary as Bayes's decision rule)

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Receiver operating characteristic (ROC) curve

- consider the binary classification case, i.e. $w, y \in \{w_1, w_2\}$. For equal losses, the optimal decision boundary is where $p(w_1|x) = p(w_2|x) = 0.5$.
- However, there are cases where unequal losses are needed, and the optimal decision boundary is no longer where $p(w_1|x) = p(w_2|x)$



- For a given threshold x_B , we can construct a confusion matrix based on the evaluation set

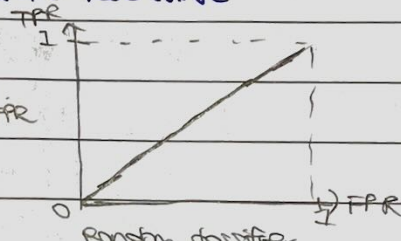
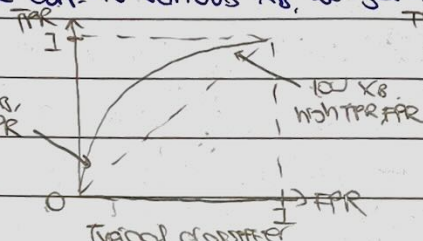
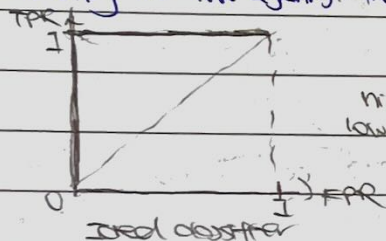
		Predicted $\hat{y} = f(x; \theta)$	
		1	0
Actual y	1	<u>TP</u>	<u>FN</u>
	0	<u>FP</u>	<u>TN</u>

- The true positive rate (TPR) and false positive rate (FPR) are defined as follows:

$$\text{TPR} = \frac{TP}{TP + FN}$$

$$\text{FPR} = \frac{FP}{FP + TN}$$

- Plotting the TPR against the FPR corr. to various x_B , we get the ROC curve



- To compare diff. ROC curves, we can consider the metric area under curve (AUC) (loses information due to dimensionality reduction)

$\Rightarrow$ Useful for determining the req. (unequal) losses and the decision boundary x_B .

Decision boundary (DB)

- The DB is a hyper-plane b/w class labels where the (log) class posteriors are equal. For the DB b/w two classes w/ labels w_1, w_2 ,

$$\log p(w_1|x; \theta) = \log p(w_2|x; \theta) \rightarrow \log p(w_1) p(x|w_1; \theta) = \log p(w_2) p(x|w_2; \theta)$$

- For a MoG model, we have multivariate Gaussian dist. as class-conditional pdfs, i.e.

$$p(x|w_i; \theta_i) = N(x|\mu_i, \Sigma_i) = \frac{1}{(2\pi)^{D/2} |\Sigma_i|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1} (x-\mu_i)\right)$$

At the DB b/w classes 1/2, $p(x|w_1; \theta_1) = p(x|w_2; \theta_2)$, which is hyper-quadratic

$$x^T A x + b^T x + c = 0$$

$$\text{where } A = \Sigma_1^{-1} - \Sigma_2^{-1}, b = 2(\Sigma_2^{-1}\mu_2 - \Sigma_1^{-1}\mu_1), c = \mu_1^T \Sigma_1^{-1} \mu_1 - \mu_2^T \Sigma_2^{-1} \mu_2 - \log\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right) - 2\log\left(\frac{p(w_1)}{p(w_2)}\right)$$

• If $\Sigma_1 = \Sigma_2 = \Sigma$, we have a linear DB, $b^T x + c = 0$

conditional independence and graphical models

Conditional independence (CI).

- If X and Y are conditionally independent given Z , i.e. $X \perp Y | Z$,

$$P(X, Y | Z) = P(X | Z) P(Y | Z)$$

- Modelling data often req. specifying a high-dim. dist. $p(X_1 \dots X_d)$, but working w/ fully flexible joint dist. is intractable.

- If we work w/ structured dist., we can use CI to rewrite $p(X_1 \dots X_d)$ as a product of simple factors evaluated only on a subset of $X_1 \dots X_d$.

↳ This results in a compact representation of the dist.

↳ This simplifies the # of the dist. parameters to data.

↳ Allows us to sum out variables efficiently (compute normalisation const. in exp's table).

- Factorisations can be encoded one-to-one as graphs (graphical models) s.t. nodes are RVs, and edges connect variables for which no CI exists.

Directed graphical models (Bayesian networks)

- A Bayesian network G is a directed acyclic graph whose nodes are RVs $X_1 \dots X_d$.

Let $PA_{X_i}^G$ be the parents of X_i in G and $ND_{X_i}^G$ be the non-descendants of X_i in G .

- G encodes the factorisation $p(X_1 \dots X_d) = \prod_{i=1}^d p(X_i | PA_{X_i}^G)$

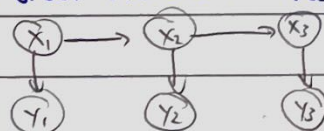
and the CI $(X_i \perp ND_{X_i}^G | PA_{X_i}^G)$, $i=1 \dots d$

- The BN expresses the joint dist. as a product of factors which depend only on a small no. of variables $\rightarrow$ we can exploit BN factorisation to avoid generating very large factors (prob. tables) during the summation process.

- Markov models are an example of BN, where we have

↳ 1st order Markov: $W_{t+1} \perp W_1 \dots W_{t-1} | W_t$

↳ 2nd order Markov: $W_{t+1} \perp W_1 \dots W_{t-2} | W_t, W_{t-1}$



$$p(X_1, X_2, X_3, Y_1, Y_2, Y_3) = p(Y_1 | X_1) p(X_2 | X_1) p(Y_2 | X_2) p(X_3 | X_2) p(Y_3 | X_3) p(X_1)$$

$$CI: Y_1 \perp X_2, X_3, Y_2, Y_3 | X_1, Y_2 \perp X_1, X_3, Y_3 | X_2, Y_3 \perp X_1, X_2, Y_1, Y_2 | X_3$$

say we want to compute $p(Y_1, Y_2, Y_3) = \sum_{X_1, X_2, X_3} p(X_1, X_2, X_3, Y_1, Y_2, Y_3) \rightarrow O(N^3)$

using CI, $p(Y_1, Y_2, Y_3) = \sum_{Y_3} \left(\sum_{X_2} \left(\sum_{X_1} p(X_1) p(X_2 | X_1) p(Y_1 | X_1) \right) p(X_3 | X_2) p(Y_2 | X_2) \right) p(Y_3 | X_3) \rightarrow O(N^3)$

* $\sum_{X_1} p(X_1) p(X_2 | X_1) p(Y_1 | X_1) = \sum_{X_1} p(X_1, X_2, Y_1) = p(X_2, Y_1)$ N entries for $X_1, X_2 \rightarrow N^2$ cost

$\sum_{X_2} p(X_2, Y_1) p(X_3 | X_2) p(Y_2 | X_2) = \sum_{X_2} p(X_2, X_3, Y_1, Y_2) = p(X_3, Y_1, Y_2)$ N entries for $X_2, X_3 \rightarrow N^2$ cost

$\sum_{X_3} p(X_3, Y_1, Y_2) p(Y_3 | X_3) = \sum_{X_3} p(X_3, Y_1, Y_2, Y_3) = p(Y_1, Y_2, Y_3)$ N entries for $X_3 \rightarrow N$ cost

Undirected graphical models (Markov networks)

- A Markov network G is an undirected graph whose nodes are the RVs $X_1 \dots X_d$.
consider positive potential functions $\phi_1(D_1) \dots \phi_k(D_k)$, where $D_1 \dots D_k$ are sets of variables, each forming a clique of G (clique is a fully-connected subset of nodes).
- G encodes the factorisation $p(X_1 \dots X_d) = \frac{1}{Z} \prod_{i=1}^k \phi_i(D_i)$ Z is the partition function,
 $Z = \sum_{x_1 \dots x_d} \prod_{i=1}^k \phi_i(D_i)$
and the CI (ALBIC) for any sets of nodes A, B, C s.t. C separates A from B in G .
- MN are better options over BN when having to choose a dir. for the edges is rather awkward (eg. symmetry in the graph).

Latent variable and sequence models

Latent variable models

Latent variable generative models

- common Bayesian networks for latent variable generative models are:

↳ Factor analysis $p(x) = \int p(x|z) p(z) dz$ $\textcircled{z} \rightarrow \textcircled{x}$

↳ Finite mixture model $p(x) = \sum_{c=1}^M p(c) p(x|c)$ $\boxed{c} \rightarrow \textcircled{x}$

↳ Discrete mixture model $p(x) = \sum_{c=1}^M p(c) p(x|c)$ $\boxed{c} \rightarrow \boxed{x}$

where x is the observation and z/c are the continuous/discrete latent variables.

(Graphically: $\square$ = discrete, $\circ$ = continuous // shaded = observed not shaded = latent)

Factor analysis (FA)

- In FA, the observations $\underline{x} \in \mathbb{R}^d$ are assumed to be generated by a process of the form

$$\underline{x} = \underline{C}\underline{z} + \underline{v}$$

where $\underline{z} \in \mathbb{R}^p \sim N(\underline{0}, \underline{I})$ is the latent variable ($p < d$), $\underline{C} \in \mathbb{R}^{d \times p}$ is the loading matrix and $\underline{v} \in \mathbb{R}^d \sim N(\underline{0}, \underline{\Sigma}_{\text{diag}})$ [$\underline{\Sigma}_{\text{diag}} = \sigma^2 \underline{I}$ is probabilistic PCA].

$$\rightarrow p(\underline{x}|\underline{z}) = N(\underline{x}|\underline{C}\underline{z}, \underline{\Sigma}_{\text{diag}}) \quad p(\underline{x}) = N(\underline{x}|\underline{0}, \underline{C}\underline{C}^T + \underline{\Sigma}_{\text{diag}})$$

Gaussian mixture model

- The Gaussian mixture model / mixture of Gaussians (MOG) model is given by

$$p(\underline{x}) = \sum_{m=1}^M \pi_m N(\underline{x}|\underline{\mu}_m, \underline{\Sigma}_m)$$

where π_m is the cluster membership distribution,

$$\pi_m = p(c=m|\theta) \quad \sum_{m=1}^M \pi_m = 1$$

Jensen's inequality

- for a concave function f , and $\lambda \in [0, 1]$.

$$\lambda f(x) + (1-\lambda)f(y) \leq f(\lambda x + (1-\lambda)y)$$

We can generalise the result to expectations

$$E[f(x)] \leq f(E[x])$$

Alternatively, for a prob distribution w_k , i.e. $\sum_k w_k = 1$,

$$\sum_k w_k f(x_k) \leq f(\sum_k w_k x_k)$$

- setting $f = \log$, we have

$$\sum_k w_k \log x_k \leq \log \sum_k w_k x_k$$

Auxiliary function Q

- We would like to optimise the model parameters θ to max. the log-likelihood $L(\theta)$.

$L(\theta)$ is difficult to max. $\rightarrow$ consider a lower bound auxiliary function $Q(\theta)$ to max.

- We want to iterate the model parameters $\theta^t \rightarrow \theta^{t+1}$ to increase the log-likelihood $L(\theta^t) \rightarrow L(\theta^{t+1})$.

$$L(\theta^{t+1}) - L(\theta^t) = \sum_{i=1}^n \log \frac{p(x^{(i)} | \theta^{t+1})}{p(x^{(i)} | \theta^t)} > 0$$

- We can rewrite $L(\theta^{t+1}) - L(\theta^t) > 0$

$$\begin{aligned} L(\theta^{t+1}) - L(\theta^t) &= \sum_{i=1}^n \log \frac{\sum_{m=1}^M p(x^{(i)}, c=m | \theta^{t+1})}{p(x^{(i)} | \theta^t)} = \sum_{i=1}^n \log \sum_{m=1}^M p(c=m | x^{(i)}, \theta^t) \frac{p(x^{(i)}, c=m | \theta^{t+1})}{p(c=m | x^{(i)}, \theta^t) p(x^{(i)} | \theta^t)} \\ &\stackrel{\text{Jensen}}{\geq} \sum_{i=1}^n \sum_{m=1}^M p(c=m | x^{(i)}, \theta^t) \log \frac{p(x^{(i)}, c=m | \theta^{t+1})}{p(c=m | x^{(i)}, \theta^t) p(x^{(i)} | \theta^t)} \\ &= \sum_{i=1}^n \sum_{m=1}^M p(c=m | x^{(i)}, \theta^t) \log \frac{p(x^{(i)}, c=m | \theta^{t+1})}{p(x^{(i)}, c=m | \theta^t)} \end{aligned}$$

We therefore have the inequality

$$L(\theta^{t+1}) - L(\theta^t) \geq Q(\theta^t, \theta^{t+1}) - Q(\theta^t, \theta^t)$$

$$\text{where } Q(\theta^t, \theta^t) = \sum_{i=1}^n \sum_{m=1}^M p(c=m | x^{(i)}, \theta^t) \log p(x^{(i)}, c=m | \theta^t)$$

$\rightarrow$ When we max. $Q(\theta^t, \theta)$ wrt θ by selecting $\theta^{t+1} = \arg \max_{\theta} Q(\theta^t, \theta)$,

$$Q(\theta^t, \theta^{t+1}) - Q(\theta^t, \theta^t) \geq 0 \Rightarrow L(\theta^{t+1}) - L(\theta^t) \geq 0$$

- For the continuous case, the auxiliary function Q becomes

$$Q(\theta^t, \theta^t) = \int p(z | x, \theta^t) \log p(z, x | \theta^t) dz$$

Negative variational free energy $F(q(z), \theta)$

- the negative variational free energy / evidence lower bound (ELBO) is a lower bound for the likelihood $\log p(x | \theta)$, and is defined to be

$$F(q(z), \theta) = \log p(x | \theta) - K(q(z) || p(z | x, \theta)) = \log p(x | \theta) - \int q(z) \log \frac{q(z)}{p(z | x, \theta)} dz$$

* KL-divergence is non-negative, so $F(q(z), \theta) \leq \log p(x | \theta)$, w/ equality iff $q(z) = p(z | x, \theta)$

- The variational free energy $F(q(z), \theta)$ can be expressed in another form. Consider $\log p(x | \theta)$.

$$\begin{aligned} \log p(x | \theta) &= \log p(x | \theta) \int q(z) dz = \int q(z) \log \frac{p(x | \theta) p(z | x, \theta)}{p(z | x, \theta)} dz = E_{q(z)} \left[\log \frac{p(x, z | \theta)}{p(z | x, \theta)} \right] \\ &= E_{q(z)} \left[\log \frac{p(x, z | \theta)}{q(z)} \right] + E_{q(z)} \left[\log \frac{q(z)}{p(z | x, \theta)} \right] = E_{q(z)} \left[\log \frac{p(x, z | \theta)}{q(z)} \right] + K(q(z) || p(z | x, \theta)) \\ \therefore F(q(z), \theta) &= E_{q(z)} \left[\log p(x, z | \theta) \right] - E_{q(z)} \left[\log q(z) \right] = \int q(z) \log p(x, z | \theta) dz - \int q(z) \log q(z) dz \end{aligned}$$

* $F(q(z), \theta) = \log p(x | \theta) - K(q(z) || p(z | x, \theta))$ is useful for the E-step (one term const. wrt $q(z)$)

$F(q(z), \theta) = \int q(z) \log p(x, z | \theta) dz - \int q(z) \log q(z) dz$ is useful for the M-step (one term const. wrt θ)

Expectation maximisation (EM) algorithm

- The EM algorithm is an iterative method to find (local) MLE of parameters θ in statistical models which involves unobserved latent variables z .
- Since the negative variational free energy $F(q(z), \theta)$ is a lower bound of the log-likelihood $\log p(x|\theta)$, optimising $F(q(z), \theta)$ gives a better and better lower bound for $\log p(x|\theta)$.
- The $(t+1)$ -th iteration of the EM algorithm is as follows:
 - ↳ (i) Initialise $q^t(z) = p(z|x, \theta^t) \rightarrow L(\theta^t) = F(q^t(z), \theta^t)$
 - ↳ (ii) M-step: For fixed $q^t(z)$, max. the lower bound $F(q^t(z), \theta)$ wrt θ

$$\theta^{t+1} = \arg \max_{\theta} F(q^t(z), \theta) = \arg \max_{\theta} \int q^t(z) \log p(x, z|\theta) dz - \int q^t(z) \log q^t(z) dz = \arg \max_{\theta} \int q^t(z) \log p(x, z|\theta) dz$$
 - ↳ (iii) E-step: For fixed θ^{t+1} , max the lower bound $F(q(z), \theta^{t+1})$ wrt $q(z)$

$$q^{t+1}(z) = \arg \max_q F(q(z), \theta^{t+1}) = \arg \max_q \log p(x|\theta^{t+1}) + \text{KL}(q(z) \| p(z|x, \theta^{t+1})) = p(z|x, \theta^{t+1})$$
- We iterate until convergence — find a local max. of likelihood (which depends on initial θ_0)
- Note that each iteration cannot decrease likelihood as

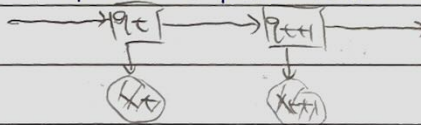
$$\log p(x|\theta^t) \stackrel{\text{E-step}}{=} F(q^t(z), \theta^t) \stackrel{\text{M-step}}{\leq} F(q^t(z), \theta^{t+1}) \stackrel{\text{lower bound}}{\leq} \log p(x|\theta^{t+1})$$
- * If we are unable to set $q(z) = p(z|x, \theta)$ (e.g. $p(z|x, \theta)$ is not a Gaussian), then

$$\log p(x|\theta^t) \geq F(q^t(z), \theta^t) \leq F(q^t(z), \theta^{t+1}) \leq \log p(x|\theta^{t+1})$$
 i.e. there is no guarantee of not decreasing the log-likelihood

Sequence models

Hidden Markov models (HMM)

- HMMs are generative models w/ dynamic Bayesian networks of the following form



$$\text{CIS: } p(q_t | q_{1:t-1}) = p(q_t | q_{t-1}) \quad ; \quad p(x_t | x_{1:t-1}, q_{1:t}) = p(x_t | q_t)$$

The joint distribution over all latent variables $q_{1:T}$ and observations $x_{1:T}$ is therefore

$$p(q_{1:T}, x_{1:T}) = \prod_{t=1}^T p(q_t | q_{t-1}) p(x_t | q_t) \quad \text{where } p(q_1 | q_0) = p(q_1)$$

The likelihood of the data can be computed by marginalising over all possible state seq. $q \in Q^T$.

$$p(x_{1:T}) = \sum_{q \in Q^T} p(q) p(x_{1:T} | q) = \sum_{q \in Q^T} \prod_{t=1}^T p(q_t | q_{t-1}) p(x_t | q_t)$$

- HMMs have emitting states that produce the observation seq. $x_{1:T}$ and non-emitting states used to define valid start and end states [N states, $s_2, \dots, s_N$ non-emitting, $s_2, \dots, s_{N-1}$ emitting]
- HMMs have the following parameters θ :

$$\begin{aligned} &\rightarrow \text{transition matrix } A \quad a_{ij} = p(q_t = s_j | q_{t-1} = s_i) \\ &\rightarrow \text{state of prob. } \{b_2(x_t), \dots, b_N(x_t)\} \quad b_j(x_t) = p(x_t | q_t = s_j) \end{aligned} \quad \left. \vphantom{\begin{aligned} &\rightarrow \text{transition matrix } A \\ &\rightarrow \text{state of prob. } \{b_2(x_t), \dots, b_N(x_t)\} \end{aligned}} \right\} \begin{array}{l} \text{usually trained} \\ \text{using EM} \end{array}$$

- HMMs can be used as a generative decoder:

$$\hat{w} = \arg \max_w p(w|x_{1:T}) = \arg \max_w p(w) p(x_{1:T} | w) \quad \left. \vphantom{\hat{w} = \arg \max_w p(w|x_{1:T})} \right\} \begin{array}{l} \text{we need to compute} \\ \text{this quickly} \end{array}$$

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Viterbi algorithm

- The Viterbi algorithm is used to find an approx (lower bound) to the likelihood $p(X_{1:T})$ and the best state-seq (path) to a state / end of entire seq.

- We approx. the likelihood as the following using $\sum_q f(q) \approx \max_q f(q)$

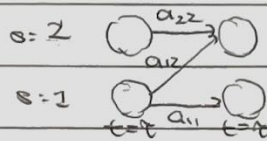
$$p(X_{1:T}) = \sum_{q \in Q^T} p(X_{1:T}, q) \approx p(X_{1:T}, \hat{q})$$

where $\hat{q}$ is the best state seq. through the discrete state space

$$\hat{q} = \{q_1, \dots, q_T\} = \arg \max_{q \in Q^T} p(X_{1:T}, q)$$

- There is an associated max log prob. for reaching state j at time t , $\phi_j(t)$.

Consider the following sub-graph



There are two paths of reaching state 2 at time t' :

↳ state 1 at time t + transition a_{12} + observation $x_{t'}$ → $\phi_1(t) + \log a_{12} + \log b_2(x_{t'})$

↳ state 2 at time t + transition a_{22} + observation $x_{t'}$ → $\phi_2(t) + \log a_{22} + \log b_2(x_{t'})$

We select the path w/ the max log prob: $\phi_2(t') = \max\{\phi_1(t) + \log a_{12}, \phi_2(t) + \log a_{22}\} + \log b_2(x_{t'})$

- The Viterbi algorithm is as follows:

↳ (i) Initialization: $\phi_1(0) = 0$, $\phi_j(0) = \text{LZERO}$ $1 < j \leq N$, $\phi_j(0) = \text{LZERO}$, $1 < j < N$ LZERO = log 0

↳ (ii) Recursion: for $t=1, \dots, T$ do

for $j=2, \dots, N-1$ do

$$\phi_j(t) = \max_{1 \leq k \leq N} \{\phi_k(t-1) + \log a_{kj}\} + \log b_j(x_t)$$

end for

end for

↳ (iii) Termination: $\log p(X_{1:T}, \hat{q}) = \max_{1 \leq k \leq N} \{\phi_k(T) + \log a_{kN}\}$

* We store all the previous best states to yield the best state seq. $\hat{q}$.

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Forward algorithm

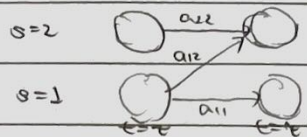
- The forward algorithm is used to find all state-seq (path) to a state

- Define the LAdd operator as follows:

$$\text{LAdd}(a, b) = \log(\exp(a) + \exp(b)) \rightarrow \exp(\text{LAdd}(a, b)) = \exp(a) + \exp(b)$$

- The total path to state j at time t is $\alpha_j(t) = \log p(x_1, \dots, x_t, q_t = s_j)$

Consider the following subgraph



There are two paths of reaching state 2 at time $t+1$:

→ state 1 at time t + transition a_{12} + observation $x_{t+1} \rightarrow \alpha_1(t) + \log a_{12} + \log b_2(x_{t+1})$

→ state 2 at time t + transition a_{22} + observation $x_{t+1} \rightarrow \alpha_2(t) + \log a_{22} + \log b_2(x_{t+1})$

The total path is the LAdd of all paths: $\alpha_2(t+1) = \text{LAdd}(\alpha_1(t) + \log a_{12}, \alpha_2(t) + \log a_{22}) + \log b_2(x_{t+1})$

- The forward algorithm is as follows:

→ (i) Initialization: $\alpha_1(0) = 0, \alpha_j(0) = \text{ZERO}, 1 \leq j \leq N, \alpha_j(0) = \text{ZERO} \quad 1 < j < N \quad \text{ZERO} = \log 0$

→ (ii) Recursion: for $t=1, \dots, T$ do

for $j=2, \dots, N-1$ do

$$\alpha_j(t) = \log\left(\sum_k \exp(\alpha_k(t-1) + \log a_{kj})\right) + \log b_j(x_t)$$

end for

end for

Forward-backward algorithm

- The forward-backward algorithm is used to find the likelihood $p(x_{1:T})$ and the posterior seq. for EM (to train the parameters θ), $p(q_t = s_j | x_{1:T})$

- The forward probability $\alpha_j(t)$ is given by

$$\alpha_j(t) = \log p(x_1, \dots, x_t, q_t = s_j) = \log\left(\sum_{k=1}^N \exp(\alpha_k(t-1) + \log a_{kj})\right) + \log b_j(x_t)$$

The backward probability $\beta_j(t)$ is given by

$$\beta_j(t) = \log p(x_{t+1}, \dots, x_T | q_t = s_j) = \log\left(\sum_{k=1}^N \exp(\beta_k(t+1) + \log a_{jk} + \log b_k(x_{t+1}))\right)$$

Add the expressions give:

$$\alpha_j(t) + \beta_j(t) = \log p(x_1, \dots, x_T, q_t = s_j)$$

- The posterior seq. for EM $p(q_t = s_j | x_{1:T})$ is given by

$$p(q_t = s_j | x_{1:T}) = \frac{1}{Z} \exp(\alpha_j(t) + \beta_j(t))$$

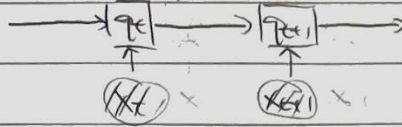
where the normalisation const Z is the likelihood $p(x_{1:T})$ $Z = \sum_{j=1}^N \exp(\alpha_j(t) + \beta_j(t)) = p(x_{1:T})$

* We just need α_j to find Z , $Z = \sum_{j=1}^N \exp(\alpha_j(T))$

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Maximum entropy Markov model (MEMM)

- MEMMs are discriminative models w/ dynamic Bayesian networks of the following form:



OIS: $p(q_t | q_{0:t-1}, x_{1:t}) = p(q_t | q_{t-1}, x_t)$

The posterior prob. of the seq. of latent variables $q_{0:T}$ (given observations $x_{1:T}$) is therefore

$$p(q_{0:T} | x_{1:T}) = \prod_{t=1}^T p(q_t | q_{t-1}, x_t)$$

where the posterior prob. of a single state q_t is given by

$$p(q_t | q_{t-1}, x_t) = \frac{1}{Z_t} \exp\left(\sum_{f \in F} \lambda_f f(q_t, q_{t-1}, x_t, t)\right)$$

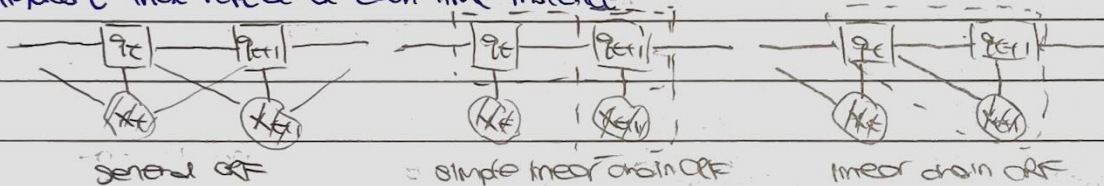
- The MEMM could be extended to the complete seq. w/ a global normalisation term

$$p(q_{0:T} | x_{1:T}) = \frac{1}{Z} \exp\left(\sum_{f \in F} \lambda_f f(q_{0:T}, x_{1:T})\right)$$

* Problem is it is diff. to handle large no. of possible features (of varying length) $\rightarrow$ many λ .

conditional random fields (CRFs).

- CRFs are discriminative models w/ dynamic Markov networks that have sets of cliques C that repeat at each time instance.



- The posterior prob. of the seq. of latent variables $q_{0:T}$ (given observations $x_{1:T}$) is given by

$$p(q_{0:T} | x_{1:T}) = \frac{1}{Z} \exp\left(\sum_{t=1}^T \sum_{C \in \mathcal{C}} \lambda_C f(q_{0:t}, x_{1:t}, t)\right)$$

where λ_C are time-independent parameters associated w/ clique C

and $f(q_{0:t}, x_{1:t}, t)$ are time-dependent features extracted from clique C w/ time-dependent label seq. $q_{0:t}$

- The simple linear chain CRF has two cliques, $C = \{C_1, C_2\}$. $C_1 = \{q_t, q_{t-1}\}$ (labeled *transition*), $C_2 = \{q_t, x_t\}$ (labeled *acoustics*).

$$p(q_{0:T} | x_{1:T}) = \frac{1}{Z} \exp\left(\sum_{t=1}^T \sum_{C \in \mathcal{C}} \lambda_C f(q_{0:t}, x_{1:t}, t)\right) = \frac{1}{Z} \exp\left(\sum_{t=1}^T \left(\sum_{i=1}^D \lambda_i^T f_i(q_t, q_{t-1}) + \sum_{i=1}^{D_0} \lambda_i^0 f_i(q_t, x_t)\right)\right)$$

the linear chain CRF has one clique $C = \{C_1\}$, $C_1 = \{q_t, q_{t-1}, x_t\}$.

$$p(q_{0:T} | x_{1:T}) = \frac{1}{Z} \exp\left(\sum_{t=1}^T \sum_{C \in \mathcal{C}} \lambda_C f(q_{0:t}, x_{1:t}, t)\right) = \frac{1}{Z} \exp\left(\sum_{t=1}^T \left(\sum_{i=1}^D \lambda_i^T f_i(q_t, q_{t-1}, x_t)\right)\right)$$

- We can use an equivalent to the forward-backward algorithm to compute the normalisation term Z .

$$Z = \sum_{q_{0:T}} \exp\left(\sum_{t=1}^T \sum_{C \in \mathcal{C}} \lambda_C f(q_{0:t}, x_{1:t}, t)\right)$$

- We train CRFs via supervised learning, w/ training observation seq. $x_{1:T}$, and label seq. $y_{1:T}$.

The model parameters λ are found via gradient descent, s.t.

$$\hat{\lambda} = \underset{\lambda}{\operatorname{argmax}} p(y_{1:T} | x_{1:T}, \lambda) = \underset{\lambda}{\operatorname{argmax}} \frac{1}{Z} \exp\left(\sum_{f \in F} \lambda_f f(x_{1:T}, y_{1:T})\right)$$

Maximum margin classifier and soft margin classifier

Constrained optimisation

- Consider the constrained optimisation problem w/ equality constraints

$$\max f(x, y) \quad \text{subject to } g(x, y) = 0$$

The sol'n (x^*, y^*) must satisfy conditions: $\nabla_{x,y} f(x, y) = -\lambda \nabla_{x,y} g(x, y)$ and $g(x, y) = 0$.

- We can find the sol'n (x^*, y^*) by optimising the Lagrangian function $L(x, y, \lambda)$ wrt x, y, λ .

$$L(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

where $\lambda \neq 0$ is the Lagrangian multiplier (could be +ve or -ve)

- Now consider the constrained optimisation problem w/ inequality constraints

$$\max f(x, y) \quad \text{subject to } g(x, y) \geq 0$$

We have two possibilities:

→ (i) Constraint is not active at the sol'n, i.e. $g(x, y) > 0$

As w/ the unconstrained case, $\nabla_{x,y} f(x, y) = 0$, and we have $\nabla_{x,y} L(x, y, \lambda) = 0$ iff $\lambda = 0$.

→ (ii) Constraint is active at the sol'n, i.e. $g(x, y) = 0$.

We have $\nabla_{x,y} f(x, y) = -\lambda \nabla_{x,y} g(x, y)$ and we req. $\lambda > 0$ (otherwise we would always set $\lambda = -\infty$)

⇒ the sol'n satisfies the Karush-Kuhn-Tucker (KKT) conditions.

$$\nabla_{x,y} f(x, y) = -\lambda \nabla_{x,y} g(x, y), \quad \lambda g(x, y) = 0, \quad \lambda \geq 0, \quad g(x, y) \geq 0$$

Binary classification problem

- Given a dataset $D = \{(x_n, t_n)\}_{n=1}^N$, formed by pairs of input features $x_n \in \mathbb{R}^d$ and target variables $t_n \in \{-1, 1\}$, we want to learn a classifier $y(x)$ s.t.

$$y(x) \geq 0 \text{ if } t_n = 1 \quad y(x) < 0 \text{ if } t_n = -1.$$

The classifier makes the correct prediction on a new i/p x_* when $y(x_*) t_* > 0$.

- The decision boundary is the set of i/p's for which $y(x) = 0$.

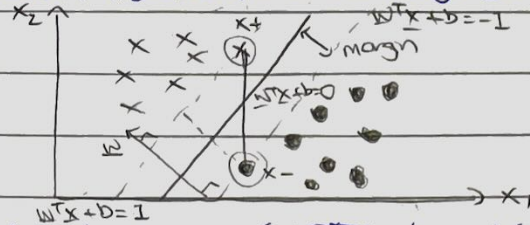
- A classification problem is linearly separable when a classifier w/ linear DB makes no mistakes on the training data. A linear classifier has the form

$$y(x) = w^T x + b$$

where w is a vector orthogonal to the DB and b is the bias.

Maximum margin classifier (MMC)

- When the data is linearly separable, many possible $\Rightarrow$ have zero training error
- The MMC is where we select a hyperplane whose distance to closest pt in each class, the margin, is maximal. The data pts on the margin are the support vectors.



- * MMCs are fully determined by its support vectors (other data pts don't affect the classifier)
- Note that the DB of linear classifiers are scale invariant, i.e. $y(x)$ and $y(x) = c y(x)$ have identical DB $\rightarrow$ we can choose the scale s.t. for the $\pm$ ve support vectors x_+/x_- ,

$$y(x_+) = w^T x_+ + b = +1$$

$$y(x_-) = w^T x_- + b = -1$$

The magnitude of the margin is therefore

$$\frac{w^T(x_+ - x_-)}{2\|w\|} = \frac{(w^T x_+ + b - w^T x_- - b)}{2\|w\|} = \frac{1}{\|w\|}$$

- $\rightarrow$ The MMC has minimal $\|w\|$ that has all data correctly classified, i.e. $t_n y(x_n) \geq 1 \forall n$.
- Since maximising $\frac{1}{\|w\|}$ is the same as minimising $\frac{1}{2}\|w\|^2$, the optimisation problem is

$$\min_{w, b} \frac{1}{2}\|w\|^2 \text{ subject to } t_n(w^T x_n + b) \geq 1 \quad n=1, \dots, N$$

i.e. we have a quadratic problem w/ linear constraints $\rightarrow$ there is a unique min.

Introducing the Lagrangian multipliers $\{a_n \geq 0\}_{n=1}^N$ we have the objective.

$$L(w, b, a) = \frac{1}{2}\|w\|^2 - \sum_{n=1}^N a_n \{t_n(w^T x_n + b) - 1\}$$

We can then consider the equivalent dual problem.

$$\max_{a_n \geq 0} \left[\min_{w, b} L(w, b, a) \right] \leftarrow \text{max. the soln to the inner min. problem}$$

For the inner minimisation problem, we req.

$$\frac{\partial L}{\partial w} L(w, b, a) = w - \sum_{n=1}^N a_n t_n x_n = 0 \quad \rightarrow \quad w = \sum_{n=1}^N a_n t_n x_n$$

$$\frac{\partial L}{\partial b} L(w, b, a) = - \sum_{n=1}^N a_n t_n = 0 \quad \rightarrow \quad \sum_{n=1}^N a_n t_n = 0$$

substituting into the expression for $L(w, b, a)$, we have

$$\max_{a_n \geq 0} \sum_{n=1}^N a_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^M a_n a_m t_n t_m x_n^T x_m \text{ subject to } \sum_{n=1}^N a_n t_n = 0$$

found via numerical methods $O(N^3) - O(N^2)$ typically $N \leq 50000$

KKT conditions req. $a_n(t_n y(x_n) - 1) = 0$, $a_n \geq 0$, $t_n y(x_n) - 1 \geq 0 \quad \forall n$.

$\therefore a_n > 0 \rightarrow t_n y(x_n) = 1$ i.e. they are support vectors.

- Let $S = \{n : a_n > 0\}$ denote the set of indices of support vectors.

The bias can be obtained from the set of constraints $\{t_n y(x_n) = 1 : n \in S\}$.

$$b = \frac{1}{|S|} \sum_{n \in S} \left\{ t_n - \sum_{m \in S} a_m t_m x_m^T x_n \right\}$$

where we average the soln for b given by each constraint for numerical stability.

- After training, predictions can be made using.

$$y(x) = \sum_{n \in S} a_n t_n x_n^T x + b$$

only the support vectors have non-zero contribution

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soft margin classifier (SMC) / support vector classifier (SVC)

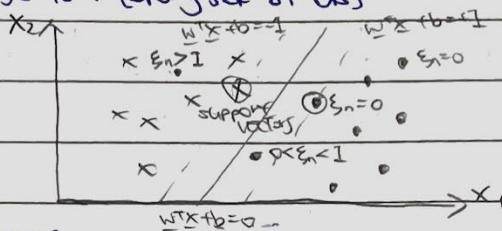
- Sometimes, allowing for mistakes in the training data produces better classifiers
 $\rightarrow$ trade-off betw margin size and no. of training errors (allow training errors $\rightarrow$ $\uparrow$ margin size)
- Errors are allowed by introducing slack variables $\{\xi_n\}_{n=1}^N$ in the constraints.

$$t_n(w^T x_n + b) \geq 1 - \xi_n, \xi_n \geq 0, n=1, \dots, N$$

Points w/ $\xi_n = 0$: correctly classified - on the margin or beyond

Points w/ $0 < \xi_n \leq 1$: lie inside the margin, but on the correct side of DB

Points w/ $\xi_n > 1$: misclassified (wrong side of DB)



- the optimisation now becomes

$$\min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{n=1}^N \xi_n \quad \text{subject to } t_n(w^T x_n + b) \geq 1 - \xi_n, \xi_n \geq 0, n=1, \dots, N$$

where $C > 0$ controls the trade-off betw the slack variable penalty and the margin

i.e. we have a quadratic problem w/ linear constraints $\rightarrow$ there is a unique min.

Introducing Lagrangian multipliers $\{\alpha_n \geq 0\}_{n=1}^N, \{\mu_n \geq 0\}_{n=1}^N$, we have the objective

$$L(w, b, \alpha, \xi, \mu) = \frac{1}{2} \|w\|^2 + C \sum_{n=1}^N \xi_n - \sum_{n=1}^N \alpha_n (t_n(w^T x_n + b) - 1 + \xi_n) - \sum_{n=1}^N \mu_n \xi_n$$

We can then consider the equivalent dual problem

$$\max_{\alpha, \mu} \left[\min_{w, b, \xi} L(w, b, \alpha, \xi, \mu) \right]$$

For the inner minimisation problem, we req.

$$\frac{\partial L}{\partial w} = w - \sum_{n=1}^N \alpha_n t_n x_n = 0 \quad \rightarrow \quad w = \sum_{n=1}^N \alpha_n t_n x_n$$

$$\frac{\partial L}{\partial b} = - \sum_{n=1}^N \alpha_n t_n = 0 \quad \rightarrow \quad \sum_{n=1}^N \alpha_n t_n = 0$$

$$\frac{\partial L}{\partial \xi_n} = C - \alpha_n - \mu_n = 0 \quad \rightarrow \quad \{\alpha_n = C - \mu_n\}_{n=1}^N$$

substituting into the expression for $L(w, b, \alpha, \xi, \mu)$, we have

$$\max_{0 \leq \alpha_n \leq C} \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m t_n t_m x_n^T x_m \quad \text{subject to } \sum_{n=1}^N \alpha_n t_n = 0$$

KKT conditions req. $\alpha_n (t_n y(x_n) - 1 + \xi_n) = 0$, $\mu_n \xi_n = 0$, $\alpha_n \geq 0$, $\mu_n \geq 0$, $t_n(w^T x_n + b) \geq 1 - \xi_n \forall n$.

$\therefore \alpha_n > 0, \mu_n > 0 \rightarrow \xi_n = 0 \rightarrow t_n y(x_n) = 1$ i.e. they are support vectors.

$\alpha_n > 0, \mu_n = 0 \rightarrow$ pt. lies within the margin boundaries

- let $M = \{n : 0 < \alpha_n < C\}$ denote the set of indices of support vectors (and $S = \{n : \alpha_n = 0\}$ is before)

The bias can be obtained from the set of constraints $\{t_n y(x_n) = 1 - \xi_n : n \in S\}$ w/ $\xi_n = 0$ (i.e. $0 < \alpha_n < C$)

$$b = \frac{1}{|M|} \sum_{n \in M} \left\{ t_n - \sum_{m \in S} \alpha_m t_m x_m^T x_n \right\}$$

where we average the sol'n for b given by each constraint for numerical stability

- After training, predictions can be made using

$$y(x) = \sum_{n \in S} \alpha_n t_n x_n^T x + b$$

Non-linear max-margin classifiers and multi-class max-margin classifiers

Non-linear max-margin classifiers / support vector machines (SVM)

- For classification problems w/ non-linear DB, we replace each feature vector $\underline{x}_n$ w/ a new one formed by applying non-linear functions to the original vector $\phi(\underline{x}_n) = \begin{bmatrix} \phi_1(\underline{x}_n) \\ \vdots \\ \phi_M(\underline{x}_n) \end{bmatrix}$
- The new classifier has a non-linear DB and has the form

$$y(\underline{x}) = \underline{w}^T \phi(\underline{x}) + b$$

The optimisation objective now becomes

$$\max_{a, b} \sum_{n=1}^N a_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N a_n a_m \phi(\underline{x}_n)^T \phi(\underline{x}_m) \quad \text{subject to} \quad \sum_{n=1}^N a_n c_n = 0$$

- For training, we only need the gram matrix $\underline{K}$, (matrix of dot products), given by.

$$K_{n,m} = \phi(\underline{x}_n)^T \phi(\underline{x}_m) \quad \Leftrightarrow \quad \underline{K} = \underline{\Phi} \underline{\Phi}^T$$

$$\text{where } \underline{\Phi} = \begin{bmatrix} \phi(\underline{x}_1) \\ \vdots \\ \phi(\underline{x}_N) \end{bmatrix}$$

After training, predictions can be made also using dot products

$$y(\underline{x}) = \sum_{n=1}^N a_n c_n \phi(\underline{x})^T \phi(\underline{x}_n) + b$$

* Cost of computing the dot product scales linearly w/ o/p dim d of feature mapping $\phi(\cdot)$

Kernel function

- so far, the gram matrix $\underline{K}$ is computed using the following two steps:

- ↳ (i) map i/p's $\underline{x}_n, \underline{x}_m$ to feature-space to obtain $\phi(\underline{x}_n)$ and $\phi(\underline{x}_m)$ (feature expansion)
- ↳ (ii) compute dot product $\phi(\underline{x}_n)^T \phi(\underline{x}_m)$

- Instead, we can use kernel function $k(\cdot, \cdot)$ to implicitly map $\underline{x}_n$ and $\underline{x}_m$ to $\phi(\underline{x}_n)^T \phi(\underline{x}_m)$.

$$K_{nm} = k(\underline{x}_n, \underline{x}_m) = \phi(\underline{x}_n)^T \phi(\underline{x}_m)$$

* $k(\cdot, \cdot)$ computes dot products in feature-space w/o explicitly performing the feature expansion.

- common kernel functions include:

$S \rightarrow 0, \underline{K} \rightarrow \text{diag} (\text{rank}(\underline{K}) = N) \rightarrow \text{wiggly DB}$
 $S \rightarrow \infty, \underline{K} \rightarrow \text{all equal entries} (\text{rank}(\underline{K}) = 1) \rightarrow \text{smooth DB}$

$$\hookrightarrow \text{Linear: } k(\underline{x}_n, \underline{x}_m) = \underline{x}_n^T \underline{x}_m$$

$$\hookrightarrow \text{Polynomial: } k(\underline{x}_n, \underline{x}_m) = (1 + \underline{x}_n^T \underline{x}_m)^d$$

$$\hookrightarrow \text{Gaussian: } k(\underline{x}_n, \underline{x}_m) = \exp\left(-\frac{1}{2s} \|\underline{x}_n - \underline{x}_m\|^2\right) \quad [\text{s controls smoothness of DB}]$$

- $k(\cdot, \cdot)$ is a valid kernel iff there is a map $\phi(\cdot)$ st. $k(\underline{x}_n, \underline{x}_m) = \phi(\underline{x}_n)^T \phi(\underline{x}_m)$ for any $\underline{x}_n, \underline{x}_m$.

In particular, iff for any data set, the gram matrix $\underline{K}$ obtained w/ $k(\cdot, \cdot)$ satisfies $\underline{K} = \underline{\Phi} \underline{\Phi}^T$.

- Mercer's condition states that $k(\cdot, \cdot)$ is a valid kernel iff the following are satisfied:

$$\hookrightarrow k(\cdot, \cdot) \text{ is symmetric, i.e. } k(\underline{x}_n, \underline{x}_m) = k(\underline{x}_m, \underline{x}_n)$$

or for any subset of data set

$$\hookrightarrow \text{Any gram matrix } \underline{K} \text{ obtained w/ } k(\cdot, \cdot) \text{ is PSD, i.e. } \underline{g}^T \underline{K} \underline{g} \geq 0 \text{ for any } \underline{g}, \{\underline{x}_{n_1}, \dots, \underline{x}_{n_N}\}$$

$$\text{- eg: } \underline{x}_n \in \mathbb{R}^2, \phi: \mathbb{R}^2 \rightarrow \mathbb{R}^5, \phi(\underline{x}_n) = [1, \sqrt{2}x_{n,1}, \sqrt{2}x_{n,2}, \sqrt{2}x_{n,1}x_{n,2}, x_{n,1}^2, x_{n,2}^2]^T$$

$$K(\underline{x}_n, \underline{x}_m) = \phi(\underline{x}_n)^T \phi(\underline{x}_m) = [1, \sqrt{2}x_{n,1}, \sqrt{2}x_{n,2}, \sqrt{2}x_{n,1}x_{n,2}, x_{n,1}^2, x_{n,2}^2] [1, \sqrt{2}x_{m,1}, \sqrt{2}x_{m,2}, \sqrt{2}x_{m,1}x_{m,2}, x_{m,1}^2, x_{m,2}^2]^T$$

$$= 1 + 2x_{n,1}x_{m,1} + 2x_{n,2}x_{m,2} + 2x_{n,1}x_{n,2}x_{m,1}x_{m,2} + x_{n,1}^2x_{m,1}^2 + x_{n,2}^2x_{m,2}^2$$

$$= (1 + x_{n,1}x_{m,1} + x_{n,2}x_{m,2})^2 = (1 + \underline{x}_n^T \underline{x}_m)^2$$

the kernel trick

- The kernel trick is that any algorithm that operates on the i/p's $\{x_n\}_{n=1}^N$ by only using dot products can be implemented using kernels

- e.g. kernelised least squares regression

The regularised least squares objective is given by

$$\text{cost}(\underline{w}, \{(\phi(x_n), t_n)\}_{n=1}^N) = \frac{1}{2}(\underline{\Phi}\underline{w} - \underline{t})^T(\underline{\Phi}\underline{w} - \underline{t}) + \frac{\lambda}{2}\underline{w}^T\underline{w} = \frac{1}{2}\underline{w}^T\underline{\Phi}^T\underline{\Phi}\underline{w} + \frac{1}{2}\underline{t}^T\underline{t} - \underline{t}^T\underline{\Phi}\underline{w} + \frac{\lambda}{2}\underline{w}^T\underline{w}$$

Differentiating wrt $\underline{w}$, we have

$$\frac{\partial}{\partial \underline{w}} \text{cost}(\underline{w}, \{(\phi(x_n), t_n)\}_{n=1}^N) = \underline{\Phi}^T\underline{\Phi}\underline{w} - \underline{t}^T\underline{\Phi} + \lambda\underline{w} = 0 \rightarrow \underline{w} = \underline{\Phi}^T(\underline{t} - \underline{\Phi}\underline{w})/\lambda = \underline{\Phi}^T\underline{a}$$

$$\Rightarrow \text{cost}(\underline{a}, \{(\phi(x_n), t_n)\}_{n=1}^N) = \frac{1}{2}\underline{a}^T\underline{\Phi}\underline{\Phi}^T\underline{\Phi}\underline{\Phi}^T\underline{a} + \frac{1}{2}\underline{t}^T\underline{t} - \underline{t}^T\underline{\Phi}\underline{\Phi}^T\underline{a} + \frac{\lambda}{2}\underline{a}^T\underline{\Phi}\underline{\Phi}^T\underline{a}$$

Differentiating wrt $\underline{a}$, we have

$$\frac{\partial}{\partial \underline{a}} \text{cost}(\underline{a}, \{(\phi(x_n), t_n)\}_{n=1}^N) = \underline{\Phi}\underline{\Phi}^T\underline{\Phi}\underline{\Phi}^T\underline{a} - \underline{\Phi}\underline{\Phi}^T\underline{t} + \lambda\underline{\Phi}\underline{\Phi}^T\underline{a} = 0 \rightarrow \underline{a} = (\underline{\Phi}\underline{\Phi}^T + \lambda\underline{I})^{-1}\underline{t}$$

predictions are then given by

$$y(\underline{x}) = \underline{w}^T\phi(\underline{x}) = \underline{a}^T\underline{\Phi}\phi(\underline{x}) = \sum_{n=1}^N a_n K(\underline{x}_n, \underline{x})$$

Multi-class max-margin classifiers

- In one-vs-all classification, we train one classifier per class.

i.e. for K classes, we train class i vs class $-i$, w/ parameters $\underline{w}_i, b_i, i=1, \dots, K$

→ we predict label for new data pt $\underline{x}^*$ using

$$\hat{c}^* = \underset{i \in \{1, \dots, K\}}{\text{argmax}} \underline{w}_i^T \underline{x}^* + b_i$$

- there are a few issues w/ one-vs-all classification:

↳ classifiers trained independently → classifiers may have diff scales → output is meaningless.

↳ Imbalanced datasets → many more training data in a particular class → bias.

↳ Some problems cannot be solved w/ linear classifiers.

- In simultaneous learning of classifiers, we add constraints to enforce o/p of correct classifier to be larger than the others.

i.e. for K classes, for each training pt $(\underline{x}_n, t_n)$ we enforce $\underline{w}_{t_n}^T \underline{x}_n + b_{t_n} > \underline{w}_j^T \underline{x}_n + b_j, j \neq t_n$.

The new objective to optimise wrt $\{\underline{w}_i\}_{i=1}^K$ and $\{b_i\}_{i=1}^K$ are

$$\min_{\underline{w}_i, b_i} \frac{1}{2} \sum_{i=1}^K \|\underline{w}_i\|^2 + C \sum_{n=1}^N \sum_{j \neq t_n} \xi_{n,j} \text{ subject to } \underline{w}_{t_n}^T \underline{x}_n + b_{t_n} > \underline{w}_j^T \underline{x}_n + b_j + 1 - \xi_{n,j}, \xi_{n,j} \geq 0 \forall n, j \neq t_n.$$

→ we predict label for new data pt $\underline{x}^*$ using

$$\hat{c}^* = \underset{i \in \{1, \dots, K\}}{\text{argmax}} \underline{w}_i^T \underline{x}^* + b_i$$

- The main problem w/ simultaneous learning of classifiers is the very large computational cost.

→ in practice, one-vs-all classifier is more widely used.

Kernels for unstructured data

Kernels for strings

- Each data pt x is a string of characters from alphabet A , i.e. $x \in A^+$, where $+$ denotes the Kleene closure operation (set of concatenating zero or more characters)

- The main idea is as follows:

→ (i): Given a list of substrings $s_1, s_2, \dots \in A^+$, each string x is encoded using the

$$\text{feature vector } \phi(x) = \begin{bmatrix} \phi_{s_1}(x) \\ \phi_{s_2}(x) \\ \vdots \end{bmatrix}$$

→ (ii): For each substring s , the feature $\phi_s(x)$ is given by the no. of occurrences of s in x ($\phi_s(x) = 0$ if s does not occur in x , o/w $\phi_s(x) > 0$)

→ (iii): The kernel b/w two strings x, x' is given by

$$K(x, x') = \phi(x)^T \phi(x') = \sum_{s \in \{s_1, s_2\}} \phi_s(x) \phi_s(x')$$

- For the gap-weighted kernel, substrings s may not need to be contiguous in x , but more gaps in the occurrence $\rightarrow$ smaller value of $\phi_s(x)$, (e.g. $\phi_s(x) = \lambda^n$, $\lambda \in (0, 1]$)

- The kernel can be computed efficiently for all possible substrings using DP

- e.g. $\phi_s(x) = \lambda^n$, $\lambda \in (0, 1]$, $s = \{ca, ct, cr, ar, rt, ba, br\}$

	ϕ_{ca}	ϕ_{ct}	ϕ_{cr}	ϕ_{ar}	ϕ_{rt}	ϕ_{ba}	ϕ_{br}
$\phi(cat)$	1	1	0	0	0	0	0
$\phi(cart)$	1	λ^2	λ	1	1	0	0
$\phi(bar)$	0	0	0	1	0	1	λ

$$\rightarrow K(cart, cart) = 1 + \lambda^2, K(cat, bar) = 0, K(cart, bar) = 1$$

- For the k-spectrum kernel, we consider all possible subsequences of length k

$\phi_k(x)$ is the no. of times that s exactly occurs in x

$K(x, x')$ computed fast using suffix tree to store all k -length seq. in x and x' ,

- We have the bag-of-words kernel when $k=1$, often applied to documents, (word strings)

- Co-occurrence of substrings is more informative for long substrings $\rightarrow k$ should be large
Common occurrences decrease as k increases $\rightarrow k$ should be small

$\Rightarrow$ we usually select intermediate values of k

- e.g. For DNA seq. we consider $s = \{AAA, AAG, \dots, TTC, TTT\}$

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Random-walk graph kernel

- For a graph $G = \{V, E\}$, V is the node set and $E = \{(i, j) : i, j \in V\}$ is the edge set.

The adjacency matrix $A \in \mathbb{R}^{N \times N}$ has entries $a_{ij} = \begin{cases} 1 & (i, j) \in E \\ 0 & \text{o/w} \end{cases}$

- A k -length graph walk is defined as $w = \{v_1, \dots, v_{k+1}\}$ where $(v_i, v_{i+1}) \in E$

The no. of k -length walks b/w nodes i and j is given by $[A^k]_{ij}$, since

$$[A^k]_{ij} = \sum_{s_1=1}^N \sum_{s_2=1}^N \dots \sum_{s_k=1}^N a_{i,s_1} a_{s_1,s_2} \dots a_{s_k,j}$$

- For the random-walk graph kernel, we have

$\phi(G)$ where the i th entry counts the no. k -length walks in G (sum of entries of A^k)

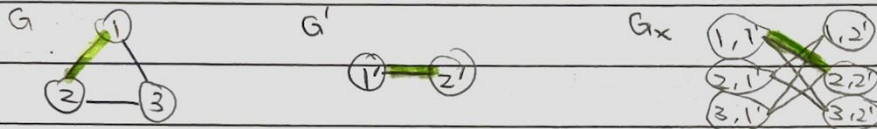
$k(G, G')$ counts the no. of common walks in graphs $G = (V, E)$ and $G' = (V', E')$

- The kernel function can be computed efficiently using the direct product graph of G and G' .

$$G_k = (V_k, E_k)$$

where $V_k = \{(a, b) : a \in V \text{ and } b \in V'\}$ and $E_k = \{((a, a'), (b, b')) : (a, b) \in E \text{ and } (a', b') \in E'\}$

- Each walk in G_k corresponds to one walk in G and G' .



- The adjacency matrix of G_k is given by $A_k = A \otimes A'$, where $\otimes$ is the Kronecker product

$$\hookrightarrow A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{p \times q} \rightarrow A \otimes B \in \mathbb{R}^{np \times mq}, \quad A \otimes B = \begin{bmatrix} a_{11}B & \dots & a_{1m}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \dots & a_{nm}B \end{bmatrix}$$

$$\text{For } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \rightarrow A \otimes B = \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{bmatrix}$$

- Assuming that $\lambda > 0$ is sufficiently small to guarantee convergence, the kernel is given by

$$k(G, G') = \sum_{i,j=1}^N \left[\sum_{n=1}^{\infty} \lambda^n A_k^n \right] = \mathbf{1}^T [I - \lambda A \otimes A']^{-1} \mathbf{1} = \mathbf{1}^T \mathbf{x}$$

where $\mathbf{x} = [I - \lambda A \otimes A']^{-1} \mathbf{1} \rightarrow$ we have $\mathbf{x} = \mathbf{1} + \lambda (A \otimes A') \mathbf{x} \rightarrow$ find $\mathbf{x}$ using the iteration.

$$\mathbf{x}_{t+1} = \mathbf{1} + \lambda (A \otimes A') \mathbf{x}_t$$

- The problem is that a walk can visit the same cycle again and again $\rightarrow$ small structural similarities can produce huge kernel values. Also it has high computational cost $O(N^3)$

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Weisfeiler-Lehman graph kernel

- The Weisfeiler-Lehman graph kernel scales well w/ large graphs and allows for node labels.

- To compute the Weisfeiler-Lehman graph kernel, we repeat for M iterations for each graph in the dataset.

↳ (i) Repeat for each node in the current graph

1) create a set w/ labels of adjacent vertices

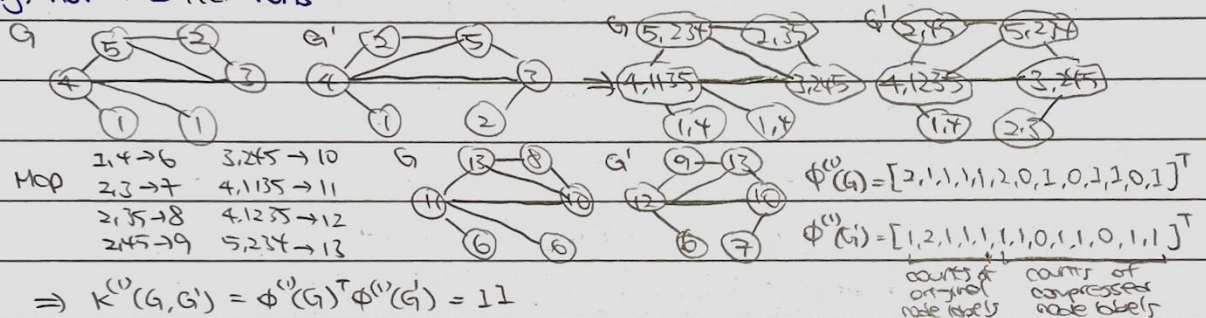
2) sort the label set and add vertex label as a prefix

3) compress resulting label rep. into a unique value

↳ (ii) Assign to each graph node its resulting unique value as a new vertex label

Then we apply the bag-of-words kernel to the vertex labels obtained throughout all the iterations (inc. the initial labels).

- e.g.: For $M=1$ iterations



Fisher kernel

- For the Fisher kernel, we use a probabilistic generative model to obtain a fixed-length vector representation of complex structured data (structure x to fixed-length score vectors)

- The Fisher kernel is computed as follows:

↳ (i) Train generative model $p(x|\theta)$ on available i/p data $\{x_n\}_{n=1}^N$, e.g. by MLE

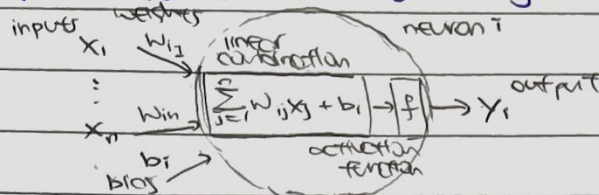
↳ (ii) Define the Fisher score vector $\phi(x_n) = \nabla_{\theta} \log p(x_n|\theta)|_{\theta=\theta_{MLE}}$

↳ (iii) The naive Fisher kernel is given by $k(x_n, x_m) = \phi(x_n)^T \phi(x_m)$

Neural network architecture

Perceptron

- The perceptron is the simplest NN architecture consisting of a single neuron

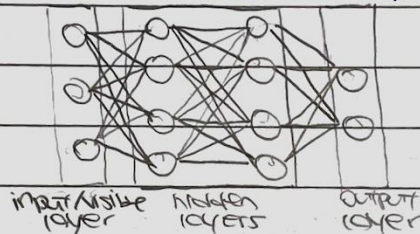


The mathematical model of the perceptron is given by

$$y_i = f\left(\sum_{j=1}^n w_{ij} x_j + b_i\right) = f(w_i^T x + b_i)$$

Multi-layer perceptron (MLP)

- MLPs (or FFNN) are a cascade of single-layer perceptrons.



Depth of network: no. of hidden layers + output layer (1: shallow; 2+: deep)

Width of layer: no. of nodes of the layer

- The width of the input layer depends on the dataset — no. of inputs
- The width of the output layer depends on the task:

↳ N-class classification: N

↳ Regression: 1

- We typically set the width of the hidden layer to be between that of the input/output layers (this can be tuned using the validation set)

- The universal approx. theorem states that a FF network w/ a linear output layer and at least one hidden layer w/ any squashing activation function can approx. any cont. function

Activation functions.

- For hidden layers, we typically use the following activation functions:

↳ ReLU (scale i/p to $[0, \infty]$)

↳ Tanh (scale i/p to $[-1, 1]$)

↳ sigmoid (scale i/p to $[0, 1]$)

- For output layers, we typically use the following activation functions:

↳ Linear (for regression)

↳ softmax (for multiclass classification)

↳ sigmoid (for binary classification)

* There are continuous equivalents to ReLU: SiLU, GeLU, ELU, SELU

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Number of output regions,

- consider a shallow neural network w/ D_i input nodes and D_h hidden nodes.

Each output consists of D_h dimensional convex polytopes.

The no. of output regions is given by

$$2^{D_i} < \sum_{j=0}^{D_i} \binom{D_h}{j} < 2^{D_h}$$

For 1D input ($D_i=1$), the no. of output regions is $1+D_h$ (no. of kinks = D_h)

- Composing two shallow neural networks is equivalent to a deep neural network, and the no. of output regions is much greater

Neural network training

Loss function

- Learning the model parameters can be cast as an optimisation problem

$$\theta^* = \underset{\theta}{\operatorname{argmin}} L(\theta)$$

where L is the loss function.

- common loss functions for regression are:

$$\hookrightarrow \text{MSE} : \frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2$$

$$\hookrightarrow \text{MAE} : \frac{1}{N} \sum_{n=1}^N |y_n - \hat{y}_n|$$

$$\hookrightarrow \text{RMSE} : \sqrt{\frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2}$$

- common loss functions for classification are:

$$\hookrightarrow \text{cross entropy} : - \sum_{c=1}^C p(y_c) \log_2(\hat{y}_c)$$

$$\hookrightarrow \text{Accuracy} : \frac{\text{correct classification}}{\text{total prediction}}$$

$\hookrightarrow \text{ROC-AUC}$

Gradient descent

- Batch size B is the no. of training samples to work through before the error is calculated and the model parameters are updated. (usually $B=2^n$, $n \in \mathbb{Z}^+$)

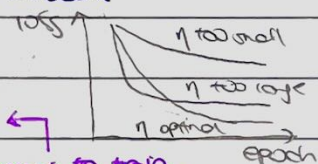
- the no. of epochs is the no. of complete passes through the training set.

one epoch consists of $\frac{\lceil \text{training set size} \rceil}{\text{batch size}}$ batches / updates.

- we can update the model parameters iteratively via gradient descent

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{1}{B} \sum_{i=1}^B \frac{dL(\theta)}{d\theta}$$

where η is the learning rate (most important hyperparameter!)



- $B=1$: stochastic gradient descent SGD

η too small: takes ages to train
may not even converge

- $B = \text{training set size}$: batch gradient descent

η too large: loss may oscillate over training epochs

- $1 < B < \text{training set size}$: mini-batch gradient descent

* Having $B < \text{training set size}$ introduces randomness $\rightarrow$ add noise, can escape local minima, less computationally expensive.

Momentum

- We can incorporate momentum into gradient descent to smooth out the updates and allow the optimization process to maintain a direction based on past gradients.
- Instead of using the current gradient, we consider a weighted sum of the current gradient and previous gradient.

$$M^{(k+1)} = \beta M^{(k)} + (1-\beta) \sum_{i=1}^S \frac{dL(\theta^{(k)})}{d\theta^{(i)}} \quad \theta^{(k+1)} = \theta^{(k)} - \eta M^{(k+1)}$$

* $\beta = 0$: gradient descent w/o momentum.

- A variation of this is Nesterov accelerated momentum, which computes the gradient a step ahead.

$$M^{(k+1)} = \beta M^{(k)} + (1-\beta) \sum_{i=1}^S \frac{dL(\theta^{(k)} - \eta M^{(k)})}{d\theta^{(i)}} \quad \theta^{(k+1)} = \theta^{(k)} - \eta M^{(k+1)}$$

Adaptive moment estimation (Adam)

- We can do gradient descent using normalised gradients as follows.

$$M^{(k+1)} = \frac{dL(\theta^{(k)})}{d\theta^{(i)}}, \quad V^{(k+1)} = \left(\frac{dL(\theta^{(k)})}{d\theta^{(i)}} \right)^2 \quad \theta^{(k+1)} = \theta^{(k)} - \eta \frac{M^{(k+1)}}{\sqrt{V^{(k+1)} + \epsilon}}$$

* Note that $\frac{M^{(k+1)}}{\sqrt{V^{(k+1)} + \epsilon}} = \text{sign}(M^{(k+1)})$ ↗ $\epsilon = 10^{-6}$ for stability

- In Adam, we update the mean and square gradients using momentum, then moderate/bias using an exponential decay $\rightarrow$ learning rate changes over time.

$$\begin{aligned} M^{(k+1)} &= \beta_1 M^{(k)} + (1-\beta_1) \sum_{i=1}^S \frac{dL(\theta^{(k)})}{d\theta^{(i)}} & \hat{M}^{(k+1)} &= \frac{M^{(k+1)}}{1-\beta_1^{k+1}} \\ V^{(k+1)} &= \beta_2 V^{(k)} + (1-\beta_2) \sum_{i=1}^S \left(\frac{dL(\theta^{(k)})}{d\theta^{(i)}} \right)^2 & \hat{V}^{(k+1)} &= \frac{V^{(k+1)}}{1-\beta_2^{k+1}} \\ \theta^{(k+1)} &= \theta^{(k)} - \eta \frac{\hat{M}^{(k+1)}}{\sqrt{\hat{V}^{(k+1)} + \epsilon}} \end{aligned}$$

- A variation of this is AMSgrad, which uses the max. of past squared gradients rather than exponential average to update the parameters $\rightarrow$ step size always increases.

$$\begin{aligned} M^{(k+1)} &= \beta_1 M^{(k)} + (1-\beta_1) \sum_{i=1}^S \frac{dL(\theta^{(k)})}{d\theta^{(i)}} \\ V^{(k+1)} &= \beta_2 V^{(k)} + (1-\beta_2) \sum_{i=1}^S \left(\frac{dL(\theta^{(k)})}{d\theta^{(i)}} \right)^2 & \hat{V}^{(k+1)} &= \max(\hat{V}^{(k)}, V^{(k)}) \\ \theta^{(k+1)} &= \theta^{(k)} - \eta \frac{\hat{M}^{(k+1)}}{\sqrt{\hat{V}^{(k+1)} + \epsilon}} \end{aligned}$$

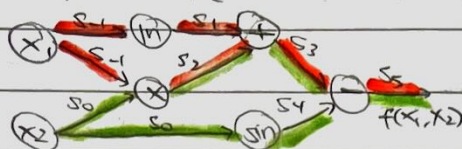
* Typically we choose $\beta_1 = 0.9$, $\beta_2 = 0.999$.

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Automatic differentiation (AutoDiff)

- We can use reverse-mode AutoDiff (backpropagation) to compute the gradients req. for gradient descent.
- In AutoDiff, the function is decomposed into a seq. of elementary arithmetic operation and functions. This can be visualised using a computational graph.

eg: $f(x_1, x_2) = \ln(x_1) + x_1 x_2 - \sin(x_2)$, $x_1 = 2$, $x_2 = 4$



$$\begin{aligned} s_1 &= x_1 & s_0 &= x_2 \\ s_1 &= \ln(s_1) & s_2 &= s_1 s_0 \\ s_3 &= s_1 + s_2 & s_4 &= \sin(s_0) \\ s_5 &= s_3 - s_4 \end{aligned}$$

Forward pass: $s_{-1} = 2$, $s_0 = 4$, $s_1 = 0.69$, $s_2 = 8$, $s_3 = 8.69$, $s_4 = -0.76$, $s_5 = 9.45$

Backward pass: $\frac{\partial s_5}{\partial s_3} = 1$, $\frac{\partial s_5}{\partial s_4} = -1$, $\frac{\partial s_3}{\partial s_1} = 1$, $\frac{\partial s_3}{\partial s_2} = 1$, $\frac{\partial s_4}{\partial s_0} = \cos(s_0) = -0.65$,

$$\frac{\partial s_1}{\partial s_{-1}} = \frac{1}{s_{-1}} = \frac{1}{2}, \quad \frac{\partial s_2}{\partial s_{-1}} = s_0 = 4, \quad \frac{\partial s_5}{\partial s_{-1}} = s_2 = 8$$

$$\rightarrow \frac{\partial f}{\partial x_1} = \frac{\partial s_5}{\partial s_1} \left(\frac{\partial s_3}{\partial s_1} + \frac{\partial s_2}{\partial s_1} \right) = 4.5, \quad \frac{\partial f}{\partial x_2} = \frac{\partial s_5}{\partial s_2} \frac{\partial s_3}{\partial s_2} \frac{\partial s_2}{\partial s_0} + \frac{\partial s_5}{\partial s_4} \frac{\partial s_4}{\partial s_0} = 2.65$$

* The forward pass propagates the function inputs down the computational graph

The backward pass computes the partial derivative of each node wrt its immediate inputs.

Feature scaling.

- Features (inputs) have varying magnitude, min-max range and units. We bring them to the same scale to improve model convergence by preventing certain features to overshadow others based solely on magnitude

- We can bring features to the same scale using normalisation or standardisation

↳ Normalisation (min-max scaling): $X_{\text{scaled},j} = \frac{x_j - \min(X)}{\max(X) - \min(X)} \in [0, 1]$

↳ Standardisation: $X_{\text{standard},j} = \frac{x_j - \text{mean}(X)}{\text{std}(X)} \in (-\infty, \infty)$

The initialisation (weight initialisation)

- consider a linear layer $y = \phi(\underline{A} \underline{x} + \underline{b})$, w/ $\underline{x} \in \mathbb{R}^N$, $\underline{A} \in \mathbb{R}^{M \times N}$, $\underline{b} \in \mathbb{R}^M$

We loosely have that $\sigma_y^2 \leq \frac{N}{2} \sigma_A^2 \sigma_x^2$, so we initialise weights w/ $N(0, \sigma_A^2 = \frac{2}{N})$

↳ For $\underline{z} = \underline{A} \underline{x} + \underline{b}$, $\sigma_z^2 = N \sigma_A^2 \sigma_x^2$; $\underline{z} = \phi(\underline{x})$, $\sigma_z^2 \leq \frac{1}{2} \sigma_x^2$

Softmax stability

- softmax is not numerically stable since $\exp(\text{number}) = \text{very large number}$.

- We consider log softmax, $\log \text{softmax}(y)_k = y_k - \log(\sum_k \exp(y_k)) = y_k - \log \text{sumexp}(y)$

where $\log \text{sumexp}(y) = \log \text{sumexp}(y - \max(y)) + \max(y)$ for stability.

* $\log \text{sumexp}(y - c) = \log(\sum_k \exp(y_k - c)) = \log(\sum_k \frac{\exp(y_k)}{\exp(c)}) = \log(\frac{1}{\exp(c)} \sum_k \exp(y_k)) = \log \text{sumexp}(y) - c$

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Batch normalisation (BN)

- BN standardises the input of each layer within a NN across the mini batch.

↳ 1) Compute the mean and variance: $\mu_B = \frac{1}{B} \sum_{i=1}^B x_i$, $\sigma_B^2 = \frac{1}{B} \sum_{i=1}^B (x_i - \mu_B)^2$

↳ 2) Standardise the input: $\hat{x} = \frac{x - \mu_B}{\sigma_B + \epsilon}$

↳ 3) Scale and shift: $z = \gamma \hat{x} + \beta$

* We introduce two learnable parameters: scale γ and shift β .

- BN has a regularising effect and allows for higher learning rates $\rightarrow$ faster training.

BN doesn't work well w/ small batch sizes or w/ seq. data (RNN)

Layer normalisation (LN)

BN/LN can be applied
before/after linear layer

- LN standardises the input of each layer within a NN across features for each data pt.

↳ 1) Compute the mean and variance: $\mu_L = \frac{1}{D_H} \sum_{i=1}^{D_H} x_i$, $\sigma_L^2 = \frac{1}{D_H} \sum_{i=1}^{D_H} (x_i - \mu_L)^2$

↳ 2) Standardise the input: $\hat{x} = \frac{x - \mu_L}{\sigma_L + \epsilon}$

↳ 3) Scale and shift: $z = \gamma \hat{x} + \beta$

* We introduce two learnable parameters: scale γ and shift β .

- LN is indep. of batch size and is effective in seq. modelling.

LN can be computationally expensive and less effective in FCN/CNN.

Vanishing gradients

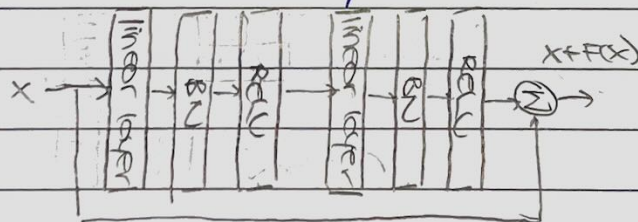
- As we add more hidden layers to a NN (i.e. increase depth), the gradients of the loss function wrt the weights become very small $\rightarrow$ weights not updated

- To mitigate the vanishing gradient problem, we can

↳ Use ReLU as the activation function - does not saturate.

↳ Use residual/skip connections - gradients flow more easily, won't become too small.

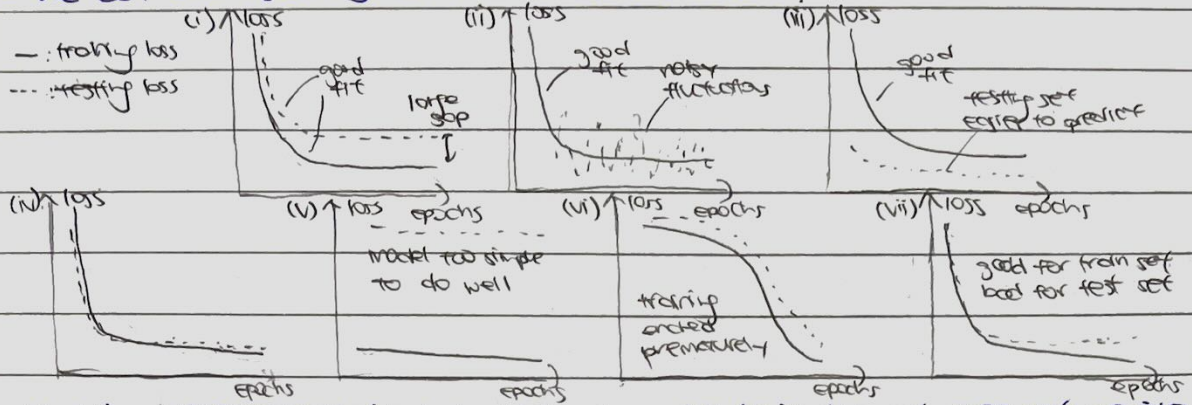
- A residual block has two branches: an identity branch in addition to the main branch



Regularisation

Learning curves

- The learning curve is a plot that shows time, steps or experience (iterations, model complexity) on the x axis and loss or accuracy on the y axis.
- We can use learning curves to spot problems.



- (i): unrepresentative train dataset — the train dataset does not provide sufficient info to learn the problem rel. to test dataset (e.g. train dataset too small)
- (ii), (iii): unrepresentative test dataset — the test dataset does not provide sufficient info to evaluate the model's ability to generalise.
- (v), (vi): model complexity too low / underfitting → ↑ model complexity, ↑ epochs
- (vii): model complexity too high / overfitting → regularisation / generalisation

Regularisation

- There may be a generalisation gap bwn the test and training loss curves due to the model overfitting or model constrained in areas w/ no training data.
- Regularisation are methods to reduce this generalisation gap. (though technically, this would be a subset of methods that add an extra term to the loss)
- common regularisation methods include:

make function smoother		increase data	
explicit L2 regularisation	apply noise to i/p	data augmentation	multi-task learning
early stopping	apply noise to o/p	transfer learning	
ensembling	dropout	apply noise to weights	implicit regularisation
combine multiple models		find wider minima	

Explicit regularisation

- In explicit regularisation, we add an extra term to the loss function,

$$\theta^* = \arg\min_{\theta} \sum_i L(y_i, \hat{y}_i) - \log p(\theta)$$

where $p(\theta)$ is a prior — gives preference to certain parameter values.

- The most common choice is L2 regularisation / Tikhonov regularisation, where smaller parameters (in magnitude) are preferred. The prior is $p(\theta) \propto \exp(-\|\theta\|^2)$, so we have

$$\theta^* = \arg\min_{\theta} \sum_i L(y_i, \hat{y}_i) + \lambda \|\theta\|^2$$

where the hyperparameter λ determines how much to penalise large $\|\theta\|$

- L2 regularisation effectively smoothens the loss surface $\rightarrow$ improved gradient descent
- * In NN, we typically only penalise the weights (not bias) — weight decay.

Implicit regularisation

- Analytical gradient descent is equivalent to solving the ODE

$$\dot{\theta} = -\eta \nabla L(\theta)$$

where $L(\theta)$ is the loss function.

- Iterative batch gradient descent results in a modified loss function

$$L_{\text{avg}}(\theta) = L(\theta) + \frac{\eta}{4} \|\nabla L(\theta)\|^2$$

encourage batches to have similar gradients

Iterative mini-batch gradient descent results in a modified loss function

$$L_{\text{avg}}(\theta) = L(\theta) + \frac{\eta}{4n} \sum_{k=1}^M \|\nabla \hat{L}_k(\theta)\|^2 = L_{\text{avg}}(\theta) + \frac{\eta}{4n} \sum_{k=1}^M \|\nabla \hat{L}_k(\theta) - \nabla L(\theta)\|^2$$

where we have M minibatches n -total and $\hat{L}_k(\theta)$ is the loss function of the k -th minibatch

- By doing gradient descent iteratively, we effectively add a regularisation term (which depends on the learning rate η)

Early stopping

- During training, the weights start small and increase in magnitude. If we stop training early, the weights don't have time to overfit to noise $\rightarrow$ reduce model complexity.
- In practice, this is implemented using a patience parameter. If the training loss does not decrease by a certain amount for more epochs than the patience parameter, training is stopped early.

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Data augmentation

- Data augmentation is a technique that enhances the diversity of training datasets w/o the need for collecting more data by creating modified versions of existing data pts.

- For image data, common data augmentation techniques include:

↳ RandomAugment: Flip, rotate, crop, stretch, blur, vignette, colour balance

↳ Mixup: Mix two or more images via a weighted sum

- In general, we can also simply add noise to the i/p or o/p.

↳ Classification: Add noise to i/p

↳ Regression: Add noise to both i/p and o/p.

// req. some physical intuition on the rough scale of i/p's or o/p's otherwise SNR too low.

Ensembling

- Ensembling is a technique that combines multiple models (taking mean/median)

- We train for a few diff. times but modify something each time (could be diff. model, diff. initialisation, diff. subsets of data resampled w/ replacement (bagging))

- Ensembling works well on parallel computers / GPUs but not great on phones as inference time is increased significantly.

Dropout

- The core idea of drop out is the randomly setting the o/p of a fraction of neurons to zero during training.

- Dropout can eliminate kinks in functions that are far from data and doesn't contribute to training loss (remove unnecessary kinks → reduce model complexity)

Transfer learning and multitask learning.

- Transfer learning is a technique where a model developed for one task is reused or adapted for a different but related task. The key steps are:

↳ 1) Pre-training: Model first trained on a large dataset for a related task.

↳ 2) Fine-tuning: pre-trained model then fine-tuned on a smaller dataset of the target task (usually modify the head and train w/ low learning rate)

- Multitask learning is an approach where a model is trained to perform multiple related tasks simultaneously - we have a trunk model and multiple head models.

* Multitask learning is diff. from transfer learning as we train the head models at the same time. (rather than have a large pre-trained model, then fine-tune the head later)

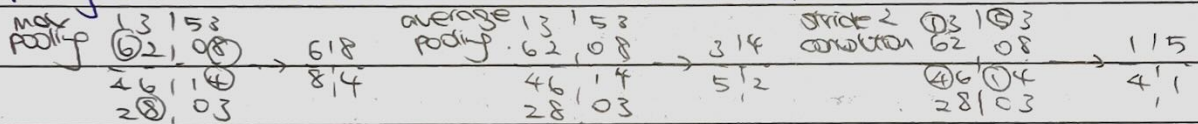
Convolutional neural network (CNN)

Convolutional layer (CONV)

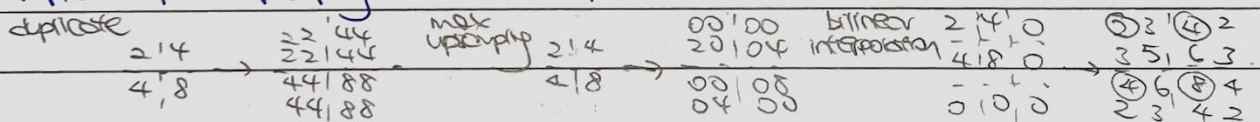
- The CONV layer applies a sliding window of filters to perform a correlation operation on the input data. (convolution is a linear operation)
- A CONV layer has the following hyperparameters:
 - ↳ Depth: no. of channels of the o/p (no. of channels of the i/p is usually fixed)
 - ↳ Kernel size: dimensions of the kernel (e.g. 3×3)
 - ↳ Stride: no. of pixels the filter moves across the i/p after each convolution operation
 - ↳ Dilation: spacing b/w kernel elements (increase receptive field w/o increasing no. parameters)
 - ↳ Zero padding: adding zeros to the edge of the image to preserve spatial dim.
- Convolution + sliding window enables translation invariance in the model.
- A CONV layer is a special case of a fully-connected layer (w/ many zero weights)
- The o/p's of the CONV layer (feature map) are typically passed through a nonlinear activation function (usually RELU) $\rightarrow$ CONV-RELU stack
 we can even linearly combine feature maps to get a single channel.
- + RELU applies the non-linearity element-wise $\rightarrow$ does not change the dimensions.

Downsampling and upsampling

- Downsampling is the process of reducing the spatial dimensions of data, typically through pooling or strided convolutions $\rightarrow$ introduce bottlenecks, extract dominant features

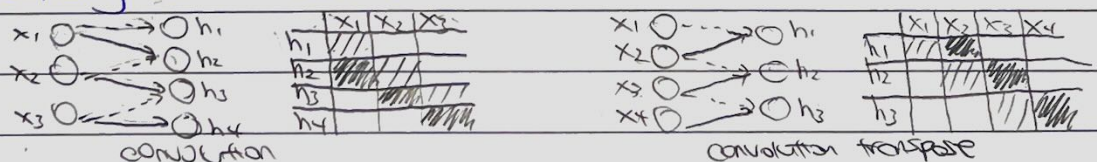


- Upsampling is the process of increasing the spatial dimensions of data, typically through duplication, max-upsampling or bilinear interpolation $\rightarrow$ restore features to original size



convolution transpose layer

- convolution transpose (deconvolution) is a layer that learns to map low-resolution feature maps back to higher dim. o/p's.
- This reverses the effects of regular convolution, effectively increasing the spatial dim while maintaining learned features



Residual network (ResNet)

- Deeper networks perform better but have the vanishing gradient problem. This can be fixed using residual / skip connections.

- The residual unit idea is rather than having a layer that transform x , we consider a layer that adds something (residual) to the previous value

$$y = x + F(x)$$

where F is the residual function (diff. b/w input x and output y)

- Each residual block layer is now computing a separate residual function w/ separate trainable parameters. (Each of them are shallow $\rightarrow$ mitigate VGP)
- * $F(x)$ and x must have the same size to perform the addition operation. (we could reduce dim. within $F(x)$, as long as we restore the dim. of the op).

Recurrent neural network (RNN)

Memory and hidden variables

- Intuitively, we can model memory by taking all the past information as input. i.e. we learn the map $\{x(t')_{t \leq t}\} \mapsto y(t)$. [Complexity $O(t^2)$]
- Instead, we can encode past information using a set of internal variables $\{h_i\}_{i=1}^n$ that represents the memory. We now learn the map $\{x(t), h_1(t), \dots, h_n(t)\} \mapsto y(t)$.
- We assume n is finite, and there exists first order dynamics to describe the evolution of the internal variables. $\dot{h}_j(t) = \phi_j(x(t), h_1(t), \dots, h_n(t)) \quad \forall j \in \{1, \dots, n\}$. [Complexity $O(t)$]

Recurrent neural network (RNN)

- Consider the case $n=1$. For a seq. of i/p $\{x(t')_{t \leq t}\}$ and o/p $\{y(t')_{t \leq t}\}$, where we know $y(t) = \psi^t(x(t')_{t \leq t})$, we can approximate ψ^t by

$$y(t) = g(x(t), h(t); \theta_g)$$

where the internal variable $h(t)$ encodes the history $\{x(t')_{t \leq t}\}$ and follows update rule

$$\dot{h}(t) = \phi(x(t), h(t))$$

Discretizing the update rule w/ forward Euler, and assuming uniform Δt .

$$\frac{h(t) - h(t-1)}{\Delta t} = \phi(x(t-1), h(t-1)) \rightarrow h(t) = f(x(t), h(t-1); \theta_f)$$

We also typically have the initial condition $h(0) = 0$.

- We define the loss of RNN to be the average loss over timespan of the i/p.

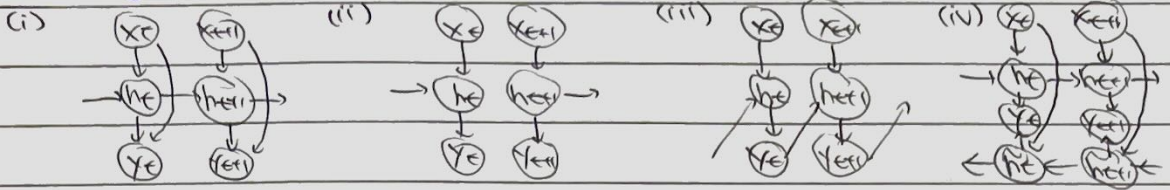
$$L(\{y(t')\}_{t \leq t}, \{\hat{y}(t')\}_{t \leq t}) = \frac{1}{t} \sum_{t'=1}^t L(y(t'), \hat{y}(t'))$$

- * A defining feature for RNNs is that θ_f and θ_g is the same for all time $t \rightarrow$ "recurrent"

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RNN variants

- We can have variants of the recurrent network



(i) original:

$$h_t = g(x_t, h_{t-1}), y_t = f(h_t)$$

(ii) Elman

$$h_t = g(x_t, h_{t-1}), y_t = f(h_t)$$

(iii) Jordan

$$h_t = g(x_t, y_{t-1}), y_t = f(h_t)$$

non-causal

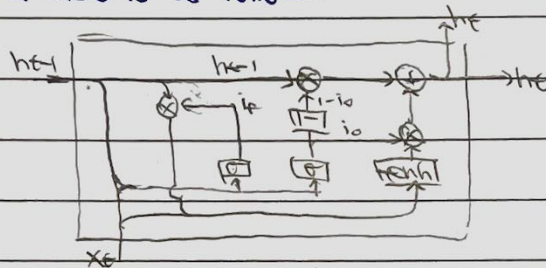
(iv) Bi-directional:

$$h_t = g(x_t, h_{t-1}), \tilde{h}_t = g(x_t, h_{t+1}), y_t = f(h_t, \tilde{h}_t)$$

- we can also add residual/skip connections, $h_t = g(x_t, h_{t-1}) + h_{t-1}$

Gated recurrent unit (GRU)

- The architecture of a GRU is as follows:



Forget gate

$$if = \sigma(w_f^x x_t + w_f^h h_{t-1} + b_f)$$

[gate over time]

output gate

$$io = \sigma(w_o^x x_t + w_o^h h_{t-1} + b_o)$$

[gate over feature (+time)]

potential hidden state

$$\tilde{h}_t = \tanh(w_h^x x_t + w_h^h (if \cdot h_{t-1}) + b_h)$$

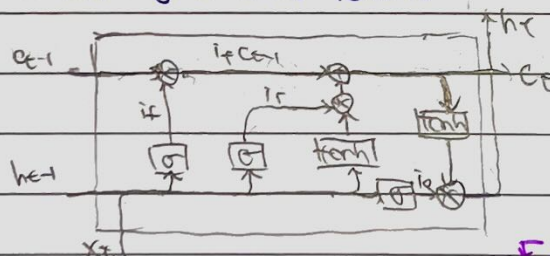
Hidden state update

$$h_t = io \cdot h_{t-1} + (1 - io) \tilde{h}_t$$

* sigmoid ops a value btwn 0 & 1.
→ useful for gating.

Long short term memory (LSTM)

- The architecture for a LSTM gate is as follows:



→ optional peephole connections

Forget gate

$$if = \sigma(w_f^x x_t + w_f^h h_{t-1} + b_f + w_f^c c_{t-1})$$

Input gate

$$ir = \sigma(w_i^x x_t + w_i^h h_{t-1} + b_i + w_i^c c_{t-1})$$

output gate

$$io = \sigma(w_o^x x_t + w_o^h h_{t-1} + b_o + w_o^c c_{t-1})$$

potential cell state

$$\tilde{c}_t = \tanh(w_c^x x_t + w_c^h h_{t-1} + b_c)$$

cell state update

$$c_t = if \cdot c_{t-1} + ir \cdot \tilde{c}_t$$

potential hidden state

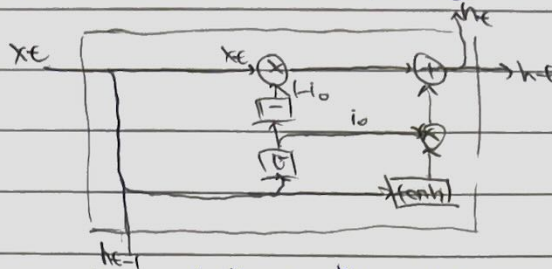
$$\tilde{h}_t = \tanh(c_t)$$

Hidden state update

$$h_t = io \cdot \tilde{h}_t$$

Highway connection

- Highway connections are similar to residual / skip connections but we have a gated contribution from the passed-through input x_t



Output gate

$$i_o = \sigma(w_o^x x_t + w_o^h h_{t-1} + b_o)$$

Potential hidden state

$$\tilde{h}_t = \tanh(w_h^x x_t + w_h^h h_{t-1} + b_h)$$

Hidden state update

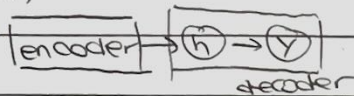
$$h_t = i_o \tilde{h}_t + (1 - i_o) x_t$$

Attention mechanism and transformers

Sequence embedding.

- Consider regression/classification of the form $y = F(X_{1:T})$ / $Y_{1:T} = F(X_{1:T})$

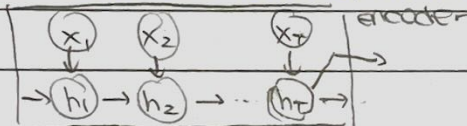
$$y = F(X_{1:T})$$



$$Y_{1:T} = F(X_{1:T})$$

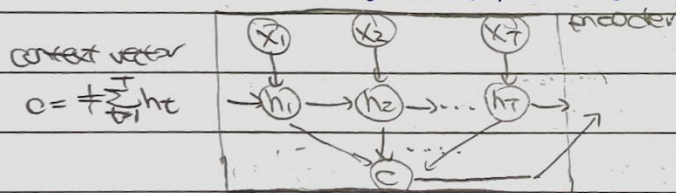


- We can use the final hidden state as information but this biases later ips.



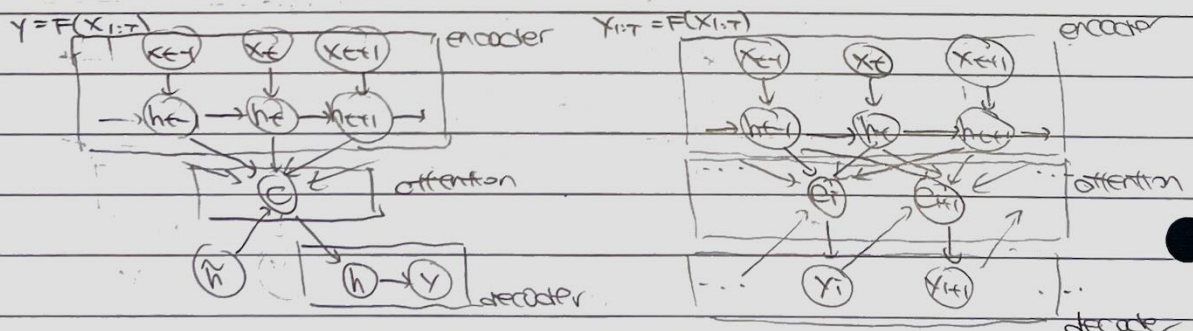
(we could use a bi-directional network, but this biases ips near the start/end)

- We can take the mean of all hidden states as information but not too effective



Instead, use weighted average which takes on input query $\tilde{h}$ - attention mechanism

(for the form $Y_{1:T} = F(X_{1:T})$, we feed the context vector c to each of the hidden states of the decoder $\rightarrow$ we req. a different decoder architecture)



- * The output y should be formed by a fixed length vector (size independent to V_p seq. length)

Attention

- The attention mechanism addresses the bottleneck problem that arises w/ the use of fixed length encoding vector where the decoder has limited access to input info
- Suppose we have an v/p o/p seq. $(\{x(t)\}_{t=1}^T, \{y(t)\}_{t=1}^L)$ where $x(t), y(t) \in \mathbb{R}^D$

The attention layer has three main parts:

- ↳ 1) Query function $W^Q \in \mathbb{R}^{d_k \times d}$ learnable weights query vector $q_j = W^Q y(t=j) \in \mathbb{R}^{d_k}$, query matrix $Q = [q_1 \dots q_L]^T$ low dim. proj. $d_k, d_v \ll n$, usually $d_k = d_v$
- ↳ 2) Key function $W^K \in \mathbb{R}^{d_k \times d}$ key vector $k_j = W^K x(t=j) \in \mathbb{R}^{d_k}$, key matrix $K = [k_1 \dots k_T]^T$
- ↳ 3) Value function $W^V \in \mathbb{R}^{d_v \times d}$ value vector $v_j = W^V x(t=j) \in \mathbb{R}^{d_v}$, value matrix $V = [v_1 \dots v_T]^T$

- We compute the similarity b/w query Q and key K using a scaled dot product.

$$\tilde{\alpha}_{ij} = \frac{q_i^T k_j}{\sqrt{d_k}} \quad \leftarrow \text{maintain appropriate variance after initialisation}$$

Then normalise using softmax (for i given T , normalise over j)

$$\alpha_{ij} = \text{softmax}(\tilde{\alpha}_{ij}) = \frac{e^{\tilde{\alpha}_{ij}}}{\sum_l e^{\tilde{\alpha}_{il}}}$$

Finally, multiply w/ the value vector to get the attention

$$y(t) = \sum_{j=1}^T \alpha_{tj} v_j$$

- In summary, we have

$$\text{attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

* In the context of seq. embedding, we have $q=h$, $k=h$, $v=h$, $Y=C \rightarrow$ we would not req. defining W^Q, W^K, W^V .

Multi-headed attention

- To extend on single-headed attention, we can have multiple heads, where each head has its own query, key, value functions.

$$\text{multihd}(\{Q, K, V\}) = \text{concat}(\text{head}_1, \dots, \text{head}_H) W^O$$

where each head, $i=1 \dots H$ is given by

$$\text{head}_i = \text{attention}(Q_i, K_i, V_i)$$

* The projection matrices W_i^Q, W_i^K, W_i^V and output matrix W^O are learnable parameters.

* For purposes of 4F10, we simply add the contributions of each head

$$y(t) = \sum_{m=1}^H \sum_{j=1}^T \alpha_{tj}^m v_j^m$$

Transformer

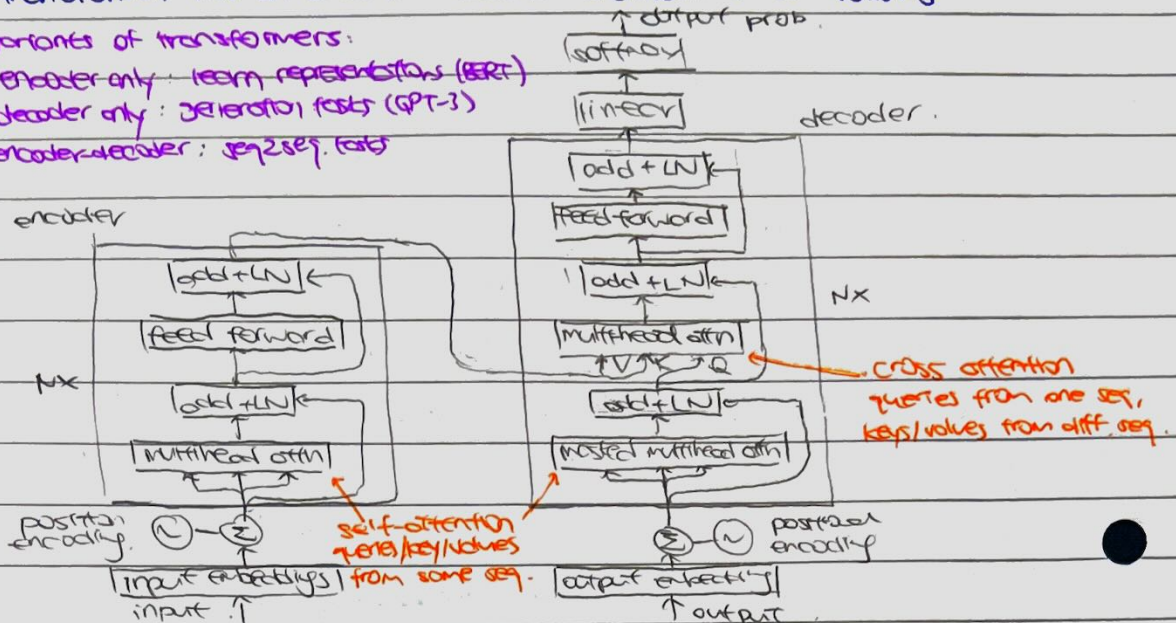
- The transformer has an encoder-decoder architecture as follows:

3 variants of transformers:

→ encoder only: term representations (BERT)

→ decoder only: generation tasks (GPT-3)

→ encoder-decoder: seq2seq tasks



- We stack N transformer blocks in both the encoder and decoder $\rightarrow N$ layers.

(Note that within each layer, there are residual connections and layer norm.)

Masked attention

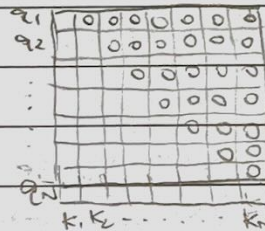
- In certain applications, we would like the transformer to attend only to

words in the past (i.e. cannot look ahead to know the answer). $\rightarrow$ "causal" attention

- This can be achieved by applying a zero mask to the attention layer.

$$(QK^T)_{ij} = 0 \quad \text{if } i > j$$

The masking pattern would look like the following:



• To ensure the rows still sum to 1, we set the top right cells to $-\infty$ before applying the softmax function.

- This is not needed for applications where it does not matter if the model can look ahead or not (e.g. translating english to french)

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Word embeddings.

not necessarily each word = token
but this is usually the case

- A sequence of words is tokenized into a seq. of integers. These can be converted into vectors using one-hot encoding.
- To obtain word embeddings, we apply an embedding matrix $A_0 \in \mathbb{R}^{N \times D}$ to the one-hot encoded vectors (practically, this is computationally wasteful $\rightarrow$ use array indexing).
- Mathematically, for an incoming sentence $W = [w_1, \dots, w_T]$, w/ one-hot encoded matrix $\tilde{W} \in \mathbb{R}^{N \times T}$, the word embeddings $X \in \mathbb{R}^{D \times T}$ are given by

$$X = A_0^T \tilde{W} = A_0[W]$$

- The embedding matrix A_0 is typically trained on general text data (e.g. word2vec, BERT), then fine-tuned on the specific target training data (transfer learning).

Positional encoding

- Attention has no representation of position, but predictions should be sensitive to location.
- We can incorporate positional encoding to our word embeddings by adding sine and cosine of diff. freq. (alternatively, this can be learned from scratch, as in ViT)

$$PE_{\tau, 2i} = \sin(\tau / 10000^{2i/D})$$

$$PE_{\tau, 2i+1} = \cos(\tau / 10000^{2i/D})$$

where τ is the position of the word, D is the no. of dim. of the word embedding.

and $2i/2i+1$ denote the index of the PE vector. $\star \text{ embedding} = \text{word embedding} + \text{PE}$

Unembedding

$A_{\text{un}} = A_0 \rightarrow$ weight tying.

- We apply the unembedding matrix $A_{\text{un}} \in \mathbb{R}^{D \times |V|}$ to the final embedding of the output, so we get a vector w/ a value for each token in the vocabulary. $X' \in \mathbb{R}^{|V|}$
- Then, we apply a softmax w/ hyperparameter temp. τ to get a prob. distribution. Including temp. allows us to control the randomness of predictions by scaling the logits before softmax.

$$\text{softmax}_{\tau}(X')_i = \frac{e^{x_i/\tau}}{\sum_j e^{x_j/\tau}}$$

$\hookrightarrow$ Increasing the temp. increases the randomness - flatter distribution

- To get an output word / seq. of words, we can either draw a single token each time or use Viterbi-style beam search to get the most likely seq.

$\langle \text{cls} \rangle$ token

- For classification tasks, we add a $\langle \text{cls} \rangle$ token at the start of the input seq. We then only take the output corr. to the $\langle \text{cls} \rangle$ token (usually we have a transformer pre-trained on general text as the trunk and have a randomly initialised MLP as the head).

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Vision transformer (ViT)

- We can use ViT model for image classification tasks. Instead of word embeddings, we have patch embeddings (w/ the first patch embedding corr. to the <cls> token).
- To generate patch tokens from an image, we divide the image into patches (eg. 16x16), then apply a linear layer on the flattened patches.
- At the output of the transformer, we have a MLP + softmax to get class probabilities.
- We can improve the performance of a ViT by distillation of a large CNN model.
 - ↳ step 1: Train a teacher CNN on ImageNet GT labels (cross entropy loss)
 - ↳ step 2: Train a student ViT to match ImageNet predictions from the teacher CNN (KL divergence loss) in addition to GT labels (cross entropy loss)
- Distillation allows the smaller student model to mimic a larger, pretrained teacher model → faster inference and reduced resource consumption on the smaller model

Image generation

Pixel RNN.

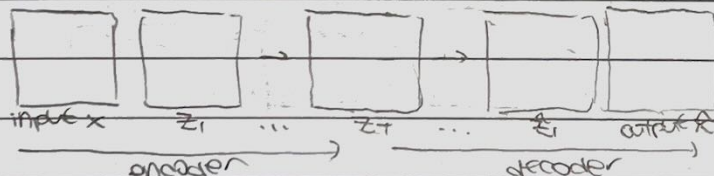
- Pixel RNN generates image pixel one at a time, starting at the top left corner.
- Each pixel depends implicitly on all pixel above and to the left. Each pixel has a hidden state that depends on the hidden state + RGB values from above + left.

$$h_{x,y} = f(h_{x-1,y}, h_{x,y-1}, w)$$

We then predict the RGB value of the pixel from its hidden state.

Diffusion models

- Consider an encoder that repeatedly adds noise to an image input (forward, diffusion process). We want to train a NN as the decoder (reverse process)



- The NN takes in a noisy image and the time step it is at, and outputs the slightly less noisy image from the previous step. (i.e. predict slightly denoised image)

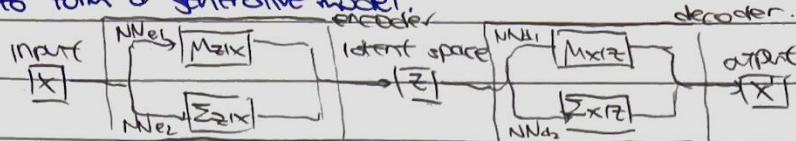
$$\hat{z}_{t-1} = \text{NN}(z_t, t; \theta)$$

- To generate an image w/ a diffusion model, we input a completely noisy image, then get a high resolution image slowly over time.

Variational autoencoders (VAEs).

- VAEs use Bayesian variational inference w/ an autoencoder (encoder-decoder)

architecture to form a generative model.



- given training images $D = \{x_i\}_{i=1}^N$, where $x_i \in \mathbb{R}^{w \times h}$, we would like to learn the probability $p(x)$ so we can draw from it (i.e. generate new image samples).
- We parameterise the prob. density by θ , and we would like to max. the log likelihood

$$\sum_{x \in D} \log p_\theta(x).$$

Consider a latent space $z \in \mathbb{R}^{d_z}$. We can rewrite the log-likelihood of a data pt using Bayes rule.

$$p_\theta(x) = \frac{p_\theta(x|z)p(z)}{p_\theta(z|x)}$$

* $p_\theta(z|x)$ is intractable $\rightarrow p_\theta(x)$ also intractable (nasty integrals)

- We set $p(z)$ to be an easy dist. to sample from, usually, the standard Gaussian $p(z) = \mathcal{N}(z|0, I)$
- Set $p_\theta(x|z)$ to have the form

$$p_\theta(x|z) = \mathcal{N}(x|M(z;\theta), \text{diag}(\Sigma(z;\theta)))$$

where the mappings $M(z;\theta): \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{w \times h}$, $\Sigma(z;\theta): \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{w \times h}$ are NNs.

- Approx. $p_\theta(z|x)$ using $q_\phi(z|x)$ of the form

$$p_\theta(z|x) \approx q_\phi(z|x) = \mathcal{N}(z|M(x;\phi), \text{diag}(\Sigma(x;\phi)))$$

where the mappings $M(x;\phi): \mathbb{R}^{w \times h} \rightarrow \mathbb{R}^{d_z}$, $\Sigma(x;\phi): \mathbb{R}^{w \times h} \rightarrow \mathbb{R}^{d_z}$ are NNs.

- The log likelihood $p_\theta(x)$ is lower bounded by the evidence lower bound (ELBO).

$$\log p_\theta(x) \geq \text{ELBO}(x;\theta,\phi) = \mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \parallel p(z))$$

Starting from $\log p_\theta(x)$,

$$\log p_\theta(x) = \log \frac{p_\theta(x|z)p(z)}{p(z|x)} = \log \frac{p_\theta(x|z)p(z)q_\phi(z|x)}{p(z|x)q_\phi(z|x)} = \log p_\theta(x|z) - \log \frac{q_\phi(z|x)}{p(z)} + \log \frac{q_\phi(z|x)}{p_\theta(z|x)}$$

Noting that $\mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x)] = \int q_\phi(z|x) \log p_\theta(x) dz = \log p_\theta(z) \int q_\phi(z|x) dz = \log p_\theta(z)$,

$$\log p_\theta(x) = \mathbb{E}_{z \sim q_\phi} [\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \parallel p(z)) + \text{KL}(q_\phi(z|x) \parallel p_\theta(z|x)) \geq \text{ELBO}(x;\theta,\phi)$$

dist. w/ prior and samples
how well q(z|x) approx p(z|x)

$\rightarrow$ We can max. $p_\theta(x)$ by max. $\text{ELBO}(x;\theta,\phi)$

- During training, we can evaluate the ELBO as follows:

① $\mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)]$ term.

$\hookrightarrow$ compute $M_{z|x}$, $\Sigma_{z|x}$ from the encoder, then sample z_s from $q_\phi(z|x)$.

$\hookrightarrow$ compute $M_{x|z}$, $\Sigma_{x|z}$ from the decoder, then compute the average of $\log p_\theta(x|z)$

② $\text{KL}(q_\phi(z|x) \parallel p(z))$ term

$$\hookrightarrow \text{KL}(q_\phi(z|x) \parallel p(z)) = \int q_\phi(z|x) \log \frac{q_\phi(z|x)}{p(z)} dz = -\frac{1}{2} \sum_{j=1}^{d_z} (1 + \log((\Sigma_{z|x})_{jj}^2) - (M_{z|x})_j^2 - (\Sigma_{z|x})_j^2)$$

- After training the VAE, we can generate data by sampling z from the prior $p(z)$, then run the sampled z through the decoder to get the dist. of x . Then we sample from the dist. of x .
- * Conditional sampling in VAEs enables more controlled (or targeted) data generation.

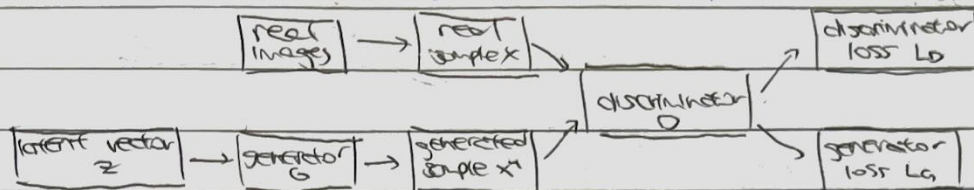
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Generative adversarial networks (GANs).

- GANs use two NNs, a generator and a discriminator, trained simultaneously through adversarial training to form a generative model.

↳ The generator learns to generate plausible data. - used as -ve training data for discriminator

↳ The discriminator learns to distinguish the generator's fake data from real data.



- The generative network $G: \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{d_x}$ takes in a latent vector $z \in \mathbb{R}^{d_z}$ (sampled from a simple dist.) and transforms it into the data space $x^* \in \mathbb{R}^{d_x}$.

- The discriminative network $D: \mathbb{R}^{d_x} \rightarrow \{0, 1\}$ is a classifier that determines whether the input is a real sample x (dps 1) or a generated sample $x^* = D(z)$ (dps 0)

- The generator and discriminator loss functions L_G and L_D are given by.

$$L_G = -L_D = \min_G \max_D E_x [\log D(x)] + E_z [\log (1 - D(x^*))]$$

→ the generator and discriminator are playing a zero sum game.